BEYOND
EVALUATION
USE

An Empirical Study of
Involvement and Use/Influence in
Multi-Site Evaluations

Report 6: A Citation Analysis of the Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education and Evaluation Fields

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Section 1: Introduction

Beyond Evaluation Use is a research project studying the use and influence of evaluations of four National Science Foundation-funded programs by examining the relationship between the extent of involvement of evaluation stakeholders and the long-term impact of the evaluations on project staff, the science, technology, engineering, and mathematics (STEM) education community, and the evaluation community. The project was funded by NSF in 2004 and is conducted by a team of researchers at the University of Minnesota.

Three program-level evaluations of NSF programs: Advanced Technological Education (ATE), Collaboratives for Excellence in Teacher Preparation (CETP), and Local Systemic Change through Teacher Enhancement (LSC), and one evaluation technical assistance project, the Math Science Partnership - Research, Evaluation and Technical Assistance project at Utah State University (MSP-RETA), form the sample used in this study[1]. Programs were selected for inclusion in this research because each of the program evaluations was designed with varying levels of involvement by the projects, with ATE being the most evaluator-driven and MSP-RETA being the most project principal investigator-driven.

This report presents findings from the effort to assess the impact of the program evaluations on the STEM education and program evaluation communities. For this study "impact" was conceptualized as the use and influence of knowledge products generated and disseminated by the program evaluations. The broad impact or influence of evaluations is a relatively new topic of interest in the evaluation community, as demonstrated in recent articles published by Mel Mark, Gary Henry, Karen Kirkhart, and Marv Alkin, and empirical investigations conducted by Carol Weiss and Tina Christie (Christie, 2007; Henry & Mark, 2003; Kirkhart, 2000; Mark & Henry, 2004; Weiss, Murphy-Graham, & Birkeland, 2005). This report presents the use of a novel approach - citation analysis - for assessing the impact of program evaluations on related fields and presents data on the extent and type of impact of these four specific program evaluations.

The following research questions guided this study:

1. To what extent have the program evaluation products (e.g., evaluation reports, journal articles, conference presentations, and evaluation instruments) been cited by individuals in the fields of STEM education and evaluation?

2. What factors appear to be related to the extent of influence these products have had on the field?

[1] For convenience sake, these four evaluation efforts will all be referred to as "program evaluations" even though the MSP-RETA project was in actuality an evaluation technical assistance project and not an overarching program evaluation.

Section 2: Methods

This study used citation analysis methods to gather data about the use of program evaluation products by individuals within the STEM education and program evaluation communities. Citation analysis is a type of bibliographic method widely used to evaluate the impact of scientific research and researchers (Hirsch, 2005). In fact, decisions about tenure and promotion within academia are often based in part on the results of citation analyses of researcher's publications. As stated on the Institute of Scientific Information's website (the developers of the Web of Science citation index):

> Citation is a direct measure of influence on the literature of a subject, and it is also a strong indicator of scientific contribution, since it is derived from patterns of interaction among millions of published articles. When one researcher cites another's work, he/she is acknowledging the relevance of that work to the current study. The interaction is both highly specific, and highly informed; it is a statement by an author of the scholarly relatedness of two works (ISI/Thomson Scientific, 2007).

Citation analysis consists of tracking the number of citations to published works. Subscription database services such as ISI's Web of Science or Elsevier's Scopus products can be used to locate citations to a particular author's work. The databases catalog citation information contained within thousands of academic journals and provide the information to subscribing institutions at a cost of up to $100,000 per year.

A recent competitor to pricey subscription services is the Google Scholar database (scholar.google.com). Google Scholar is available online and is free and easy to use; however, it does not provide a public listing of what is indexed in the database. A number of empirical studies have found that it indexes a greater amount of grey literature (i.e., papers not published in academic journals such as conference presentations, preprint or unpublished manuscripts, books, newsletters, etc.), than other citation databases. In some disciplines, however, Google Scholar's coverage of academic sources has been found to be less complete than its subscription competitors (Schroeder, 2007; Yang & Meho, 2006).

The general Google search engine (www.google.com) is a third mechanism for identifying references, although it is not a formal citation database. Its search function can be used to find references embedded in the grey literature as well as references posted in even less formal ways such as on organizational or personal websites.

This study is, to our knowledge, the first effort to use citation analysis to assess the impact of products produced by program evaluations.

Citation Analysis Methodology Employed in this Study

In this study, Web of Science, Google Scholar, and Google were used to collect citation information from lists of evaluation products produced by the four program evaluations of interest (see Appendices A-E for lists of the evaluation products produced by the four program evaluations and citations to the products). The lists of program evaluation products were

compiled for each program evaluation by reviewing the principal investigators' curriculum vitae, program evaluation archives, and through Internet web searches. Products were defined as any publications, presentations, instruments, or other related materials that were produced as part of the NSF-funded program evaluations. Additional publications that were substantially informed by the principal investigator's experiences conducting the program evaluation were identified by the PIs themselves and added to the study. In addition, we provided each principal investigator with the opportunity to review the final lists for completeness and accuracy.

Searching was completed between the following dates for each evaluation:

Table 1. Citation search dates

Evaluation	Web of Science Searching	Google Scholar Searching	Google Searching
ATE	August 27-30, 2007	September 4-6, 2007	September 6 and 10, 2007
CETP	August 16 and 18, 2007	August 19-23, 2007	December 3-4, 2007
LSC	October 23-25, 2007	October 25-November 13, 2007	November 13-December 2, 2007
MSP-RETA	September 25, 2007	September 25-26, 2007	September 26, 2007

Data were collected manually by entering full reference information about the citations into an Excel spreadsheet, which was later checked for accuracy and compliance with APA style and converted into an SPSS data file. Following data cleaning, the primary researcher for this study coded each product according to their fields and subfields. The intra rater reliability of the coding process was checked by recoding a random sample of 25 products. The percentage of agreement for the general field codes was 88% and 100% for the sub-field codes. A few miscoded items in the original data were discovered through this process and the definitions were made more specific so as to provide clear information about the meaning of the codes. To establish the level of inter-rater reliability, an outside researcher was trained in the coding scheme and was given the same 25 randomly selected products to code. However, the second coder did not have the same level of knowledge about the program evaluations and also did not apply the codes strictly according to the coding sheet. Despite these factors, the agreement levels were still reasonably high providing additional confidence in the original coding. The full coding guidelines are presented in Appendix F to clarify the categorization scheme used in this study.

Web of Science Search Process

Access to the ISI Web of Science was granted through the University of Minnesota library system. The database was searched using the cited reference search function which searches for matches in the reference lists of all the journal articles indexed in the database. The database was searched for the names of all of the authors of the evaluation products within time periods appropriate to each specific program evaluation. Additionally, searches were conducted for works authored by the National Science Foundation, Advanced Technological Education, Collaboratives for Excellence in Teacher Preparation, Evaluation Center, Horizon Research, Local Systemic Change, Math Science Partnership, Utah State University, and related variants. Appendix G contains the full list of search terms used in these searches.

The Web of Science search results in abbreviated form often did not contain sufficient information to determine whether a particular reference was actually the product of interest. The database might report that there was a citation to an article by "F Lawrenz" published in 2004 but the title of the work was abbreviated as "sci ed eval" or something similarly too vague to ascertain whether the work was a CETP product or another article by Lawrenz. In these cases, the researcher obtained and reviewed reference lists from the original sources (e.g., original journal articles) to ensure accuracy in collecting citations.

Google Scholar Search Process

The Google Scholar searches were conducted on the same lists of author names as were used in the Web of Science search, as well as by searching for exact product titles and likely variants. Combinations of author names and titles were also used to refine searches as necessary. As with the Web of Science searches, it was necessary for the researcher to obtain access to original reference lists to verify the accuracy of citations in many cases. Additionally, the researcher had to conduct research to verify the reference information of the citing works, as it was found that Google Scholar's reporting of title, author names, and sources was sometimes inaccurate.

Google Search Process

In many cases, author name searches were unmanageable using the Google search engine. For example, searching for "Iris Weiss" (in quotations) returned 14,500 results. Consequently, while author name searches were attempted, title searching was used as the primary method for culling through the numerous possible results for actual citations. As mentioned above, much time was needed to verify whether the returned results were true citations to the evaluation products of interest. Again, reference lists from the original citing documents were consulted when available to verify whether the returned results were actual citations as well as to collect the necessary data to write a full reference of the citing work.

Accuracy and Limitations

The accuracy of the citation data and method were examined by having two researchers attempt to replicate the original citation data results with a random sample of 25 evaluation products from the four program evaluations. The researchers were the original primary researcher and one other member of the Beyond Evaluation Use project team who was trained by the original primary researcher. The researchers were blind to the original findings when collecting the citation information. The replication by the primary researcher was conducted from January 14-16, 2008. The second researcher conducted her replication between January 14 and February 19, 2008. Consequently, the replication study took place between five months and six weeks after the original data collection period, depending on the particular products.

As shown in Table 2, the total number of citations found for the 25 products using each search engine during the collection trials was the same for Web of Science, but greater for both Google Scholar and Google for the replication trials conducted at later time points.

Table 2. Number of citations found for original and replication studies

Search Engine	Original Study	Replication by Primary Researcher	Replication by Secondary Researcher	Total (combining all three trials)
Web of Science	3	3	3	4
Google Scholar	19	25	26	36
Google	10	21	15	27
Total (Unique Citations)	*27*	*37*	*34*	*47*

Examining differences in the mean number of citations per product using paired t-tests, significant differences were found between Web of Science and both Google and Google Scholar (t=-3.57, df=24, p=0.002 and t=-3.572, df=24, p=0.002), however the observed differences between Google and Google Scholar were not statistically significant (Table 3).

Table 3. Mean citations per product – different databases

	Mean Citations per Product	N	SD
Web of Science	.16	25	.473
Google	1.08	25	1.579
Google Scholar	1.44	25	1.873

As shown in Table 4, the mean number of citations per evaluation product differed between the three data collection trials. Analyzing the data using paired t-tests, a significant difference in the mean number of citations per products was found only between the original data and the replication by the primary researcher (t=-2.449, df=24, p=.022). Neither the differences observed between the original and secondary researcher replication nor the differences between the primary and secondary researchers during the replication study were found to be statistically significant.

Table 4. Mean citations per product – different data collection trials

	Mean Citations per Product	N	SD
Original Study	1.08	25	1.778
Primary Researcher Replication	1.48	25	2.124
Secondary Researcher Replication	1.36	25	1.868

In addition to examining citation counts, the percent agreement of the citations between the three trials was calculated. As shown in Table 5, the percent agreement for the Web of Science data

was extremely high between each of the trials, while the Google Scholar and Google results have lower agreement levels due to the greater number of citations found during the replication studies.

Table 5. Percent agreement of citation information

Search Engine	Original and Replication by Primary Researcher	Original and Replication by Secondary Researcher	Replications by Primary and Secondary Researchers	All Three Trials
Web of Science	96.6%	96.6%	100.0%	96.6%
Google Scholar	72.9%	64.4%	78.0%	57.6%
Google	67.8%	78%	76.3%	61.0%

Factors affecting the consistency between data collection trials. There are several factors that may help to explain the discrepancies between the data collected from the different trials. First, it is clear that time is a factor in collecting citation information: every day new works are being published with new reference lists and, at some regular (and unknown) interval, citation search engines are being updated and expanded to include this additional content.

Second, products that were produced a longer time ago have a greater opportunity to be cited than those produced more recently. As shown in Table 6, although 50% of citations occurred within first two years after product publication. We found that there is, on average, three years of difference between the publication date of a product and its subsequent citation.

Table 6. Years between publication date and citation date

Difference in Years	N	%
0	52	15.0
1	64	18.4
2	59	17.0
3	51	14.7
4	30	8.6
5	19	5.5
6	28	8.1
7	20	5.8
8	15	4.3
9	6	1.7
10	3	.9
Total	347	100.0

While the first time consideration affects the data collected on each of the four program evaluations in theoretically similar ways, it is important to acknowledge the second time factor as it differentially affects the program evaluations' opportunities to publish and be cited. For example, the LSC core evaluation began in 1995, four years earlier than the ATE evaluation, which began in 1999. The CETP core evaluation began in 1999, and the Utah State RETA

project was started in 2002. Therefore, while many of the citations to the products may have already occurred for some of the products, we would expect many more to come in the future, especially for MSP-RETA.

A second major factor that emerges from the data presented above is that the particular researcher collecting the data has an effect on what is found. While the data found using Web of Science was consistent between researchers at the same point in time, the data found using Google Scholar and Google vary between the two researchers. This highlights that there is some art as well as science in collecting citation information. It is interesting, however, that in terms of overall mean citation counts it is only the differences between the primary researchers first and second data collection attempts that significant differences are found. This suggests that researcher experience is an important factor in the number of citations found during a particular search. While the secondary researcher was trained in the primary researcher's search process, there is a skill constructing the best search terms, refining search results, and in capturing accurate data. While it is hard to assess the skill levels of each of the collectors, in terms of the amount of time it took for each researcher to conduct the replication study there were large differences. It took the original researcher approximately 12 hours to collect citation data on the 25 products while it took the secondary researcher approximately 27 hours to do the same task.

In the case of collecting data using the Google search engine, there is an additional complicating factor that may contribute to differences in the data that are obtained. Google search results are tailored based on what Google knows about a particular user[2]. For example when a user searches for Jaguar, the results returned with links to either cars or cats will appear in different orders depending on the user. Consequently, the results returned for a particular search may have been more accurate for the primary researcher during the replication study as Google had already "learned" her search preferences during the original data collection period.

A final source of error is that it is harder to find citations to certain types of evaluation products than to others. In particular, finding citations to instruments is difficult. Some authors appear not to include references to instruments formally in their reference lists. As an example, the dissertation: Sciulli, J.A. (2004). Teaching Science Through Inquiry in K-5 Classrooms: Analysis of Change in Practice. Doctoral dissertation: Duquesne University. The dissertation includes a letter from Iris Weiss, PI of the LSC program evaluation, stating that he can use her questionnaire if recognition is given. However, there is no reference to the instrument in the reference list – just a mention in the text itself and the inclusion of the letter. This more informal way of citing instruments makes it harder to capture them using the search tools, in particular, Web of Science and Google Scholar, which index reference lists.

In conclusion, the citations that were found in this study should be seen as only a cross-sectional sample of the "true" citations to the works. The results are bound by the time frame during which they were collected and shaped by the specific researcher who collected the data. That said, the research was conducted in as accurate and systematic a way as possible and the citations collected are, to the best of our knowledge, a representative sample of actual citations. As a research team we feel that while this methodology may have some limitations, the data collected provide one useful snapshot of the influence of these evaluations on the STEM education and evaluation fields.

[2] Per personal conversation with Warren Teitelman, Google software engineer, 10/20/07.

Section 3: Findings

This section of the report presents the findings from the citation analysis of the 246 products produced by the four evaluations. The findings are reported without making mention to whether they were obtained through the Web of Science, Google Scholar, or Google search engine. Discussion of differences in the results obtained through the three search engines, as well as other technical and validity aspects of citation analysis, is explored in more detail in: Greenseid, L.O. (in production). *Measuring STEM Education Evaluation Influence using Citation Analysis Methods.* Doctoral Dissertation: University of Minnesota.

Section 3.1: Description of ATE, CETP, LSC, and MSP-RETA Products, Citations, and Citing Works

This section begins by describing the 246 products produced by the four evaluations and the citations they received, followed by a description of the 280 unique works that cited one or more of the evaluation products.

Descriptions of the Evaluation Products

Between the four evaluations a total of 246 evaluation products were produced through the end of 2006, including one product that was informed by three of the four evaluations (ATE, CETP, and LSC). As shown in Table 7, the different evaluations produced different quantities and types of products. It is not surprising that the ATE and LSC evaluations produced the most products, as they were funded at higher levels and for longer time frames. Also, in their last years of funding, both the ATE and LSC evaluations emphasized dissemination of findings to the field. Both ATE and LSC produced a total of 98 products each. CETP and MSP-RETA produced 24 and 25 products respectively.

When examining the bivariate correlations between program evaluation (ATE, CETP, LSC, MSP-RETA) and number of citations, significant differences were found in the average number of citations received by products produced by the four program evaluations, excluding the multi-program product ($F=2.985$, $df=3$, $p=.032$). It should be noted, however, that these differences account for only 3.6% of the variance in mean citation counts and, as discussed later, program evaluation is not significant in multiple regression analyses.

As shown in Table 7, the mean number of citations per product for each of the evaluations ranged from .4 to 2.53. Interestingly, however, post-hoc analyses of differences between the evaluations showed that the only significant difference in mean product citation counts was between the ATE evaluation (mean=.65) and the LSC core evaluation (mean=2.53).

Table 7. Program evaluation products and citation counts

Program evaluation	Number of Products	Number of Citations	Mean Citations/ Product
ATE	98	64	.65
CETP	24	42	1.75
LSC	98	248	2.53
MSP-RETA	25	10	.40
Total	245*	364	1.49

*N=245 as one product produced by multiple evaluations was not included

Product types. Evaluation products were coded by type according to the following typology:

- Instrument/Tools: Evaluation instruments (observation protocols, surveys, etc.); instrument manuals, online evaluation tools (logic model tool, resource database);
- Presentations: Conference presentations, presentations to NSF or other audiences;
- Publications: Journal articles, books, book chapters, published monographs, dissertations, newsletters;
- Reports: Evaluation reports, briefing papers, fact sheets.

Evaluation reports accounted for almost 50% of the products produced by the four program evaluations in aggregate, however within each evaluation the distribution varied. Over half of the products ATE produced were evaluation reports (n=52), while presentations (n=27) and publications (n=19) comprise the remaining half. ATE did not publish any evaluation instruments or tools. CETP produced almost equal numbers of instruments (n=5), reports (n=5), and publications (n=4) and a number of presentations as well (n=10). LSC produced 42 evaluation reports and 33 publications. The remaining products produced by LSC were instruments (n=13) and presentations (n=10). MSP-RETA focused on presentations (n=18). The remaining quarter of MSP-RETA products were reports (n=4), instruments (n=2), and publications (n=1). The one multi-evaluation product was a publication.

Table 8. Evaluation product types

Program Evaluation		Evaluation Product Type				Total
		Instrument/Tool	Presentation	Publication	Report	
ATE	N	0	27	19	52	98
	%	.0%	27.6%	19.4%	53.1%	100.0%
CETP	N	5	10	4	5	24
	%	20.8%	41.7%	16.7%	20.8%	100.0%
LSC	N	13	10	33	42	98
	%	13.3%	10.2%	33.7%	42.9%	100.0%
MSP-RETA	N	2	18	1	4	25
	%	8.0%	72.0%	4.0%	16.0%	100.0%
Multiple	N	0	0	1	0	1
	%	.0%	.0%	100.0%	.0%	100.0%
Total	N	20	65	58	103	246
	%	8.1%	26.4%	23.6%	41.9%	100.0%

Differences by product type. Significant differences were found in the mean number of citations for different types of evaluation products ($F=13.241$, $df=3$, $p<.001$). The differences in product type accounted for 14.1% of the variance in mean citations. As shown in Table 9, evaluation instruments and tools were cited most frequently, averaging 7.25 citations per product. This was

significantly higher than the other three product types which ranged between zero citations per presentation to 1.72 citations per report.

Table 9. Citations for types of products

Evaluation Product Type	Number of Products	Mean Citations	Number of Citations
Instrument/Tool	20	7.25	145
Report	103	1.72	177
Publication	58	.95	55
Presentation	65	.00	0
Total	246	1.53	377

Product fields. In addition to being coded by type, products were coded as to the field and sub-field they represented[3]. Product fields were broadly defined according to the following:

- Education–general: education areas/topics outside of STEM education;
- Evaluation-general: evaluation areas/issues outside of STEM evaluation;
- STEM education/research: general topics or research related to STEM education;
- STEM evaluation: topics or findings connected to a specific STEM evaluation;
- Other: not falling into one of the above categories.

As shown in Table 10, as to be expected, the vast majority of evaluation products were related to STEM evaluation, meaning that they were evaluation reports or articles/presentations related to one of the four specific program evaluations. ATE had a large proportion of general evaluation related products and the only general education product among the group. CETP had a couple general evaluation and STEM education/research related products. LSC had a number of STEM education/research type products in addition to STEM evaluation products. MSP-RETA had a few general evaluation products but no general or STEM education products.

[3] See Appendix F for more detailed coding guidelines.

Table 10. Fields of evaluation products

Program Evaluation		Field of Evaluation Product: Education - general	Evaluation - general	STEM education / research	STEM evaluation	Total
ATE	N	1	18	3	76	98
	%	1.0%	18.4%	3.1%	77.6%	100.0%
CETP	N	0	2	2	20	24
	%	.0%	8.3%	8.3%	83.3%	100.0%
LSC	N	0	1	14	83	98
	%	.0%	1.0%	14.3%	84.7%	100.0%
MSP-RETA	N	0	5	0	20	25
	%	.0%	20.0%	.0%	80.0%	100.0%
Multiple	N	0	1	0	0	1
	%	.0%	100.0%	.0%	.0%	100.0%
Total	N	1	27	19	199	246
	%	.4%	11.0%	7.7%	80.9%	100.0%

Product sub-fields. Products were coded as to the specific sub-field within education or evaluation according to the following[4]:

- Professional development: teacher training during service including the effects of PD on student achievement, etc.;
- Workforce development: high tech, STEM workforce training/development;
- Partnerships/systemic reform: university, school, community partnerships and systemic reform initiatives;
- Curriculum/instruction/assessment: related to classroom teaching: (i.e., instruction methods, curriculum, pedagogy, student assessment, standards-based instruction, math/science reform);
- Evaluation: evaluation-related topics such as multi-site evaluation theory, evaluation instrument protocols/manuals, other general evaluation theory or knowledge;
- Materials development: development of instructional materials;
- Project capacity building: issues related to program improvement, sustainability, advisory committee development, dissemination, replication;
- Other: other fields.

[4] See Appendix F for further details regarding these categories. Please note that evaluation instruments, manuals, and protocols were coded as "evaluation" rather than what topic the instrument collected data about.

The specific fields of the evaluation products varied according to the focus of each of the programs. ATE, where the primary goal of the program is to develop a high-tech workforce, had about a third of its products related to that topic (n=32). ATE also produced a high number of evaluation related products (n=38). There are also a number of project capacity building, workforce development and professional development products. The CETP evaluation products were predominantly related to evaluation topics and instruments (n=19). A few products specifically related to issues in pre-service teacher training and curriculum/instruction/assessment were also produced. LSC predominantly produced products related to issues in professional development (n=72), also reflecting the emphasis of the LSC program. The MSP-RETA project produced a number of evaluation-related products (n=19) and products addressing issues in partnerships (n=6).

Table 11. Specific field of evaluation products

Program evaluation		Specific Field of Evaluation Product								Total
		Teacher training/ Pre-service	Prof. Dev.	Workforce Dev.	Partner-ships/ systemic reform	Curriculum/ Instruction/ Assessment	Evaluation	Materials Dev.	Project Capacity Building	
ATE	N	0	5	32	1	2	38	6	14	98
	%	.0%	5.1%	32.7%	1.0%	2.0%	38.8%	6.1%	14.3%	100.0%
CETP	N	3	0	0	0	2	19	0	0	24
	%	12.5%	.0%	.0%	.0%	8.3%	79.2%	.0%	.0%	100.0%
LSC	N	1	72	0	3	5	17	0	0	98
	%	1.0%	73.5%	.0%	3.1%	5.1%	17.3%	.0%	.0%	100.0%
MSP-RETA	N	0	0	0	6	0	19	0	0	25
	%	.0%	.0%	.0%	24.0%	.0%	76.0%	.0%	.0%	100.0%
Multi	N	0	0	0	0	0	1	0	0	1
	%	.0%	.0%	.0%	.0%	.0%	100.0%	.0%	.0%	100.0%
Total	N	4	77	32	10	9	94	6	14	246
	%	1.6%	31.3%	13.0%	4.1%	3.7%	38.2%	2.4%	5.7%	100.0%

Field and subfield citation counts. Significant differences were not found in the mean number of citations received by products of different fields and subfields. On average, products of different fields and subfields each received 1.53 citations.

Description of the Works Citing the Evaluation Products

A total of 377 references were found to the 246 evaluation products. The 377 references were found in 280 unique citing works (see Appendix H for details regarding each of the unique citing works). While the vast majority of the citing works only cited one product (n=223), 39 of the citing works cited two evaluation products, eleven cited three products, and the remaining seven cited more than three products. One citing work cited fourteen of the products examined in this study. This citing work was the capstone report produced by the LSC core evaluation team (Banilower et al., 2006).

Table 12. Number of references to evaluation products by citing works

Number of references to evaluation products	Number of citing works	%
1	223	79.6
2	39	14.0
3	11	3.9
4	3	1.1
5	2	.7
7	1	.4
14	1	.4
Total	280	100.0

Citing works – author descriptions. As shown in Table 13, the majority of citations (56%) came from sources not affiliated[5] with the four evaluations in some way (coded as "Other"). The remaining 44% were related to the evaluations in some way; the authors were PIs or staff of one of the four evaluations (coded as the name of the evaluation if the citing work was produced by that evaluation, or coded as "PI" or "staff" if the author was writing in a different role); one of the projects funded by the ATE, CETP, LSC or MSP programs; a different NSF funded project or program ("NSF Other Project") or a NSF authored document ("NSF"). An appendix with specific author names (individuals or organizations) is included in Appendix I.

Table 13. Authors of citing works

Citing Author	N	%
Other	156	55.7
CETP Project	21	7.5
LSC Core	21	7.5
ATE Evaluation	18	6.4
LSC Project	17	6.1
NSF Other Project	10	3.6
PI	10	3.6
CETP Core	8	2.9
NSF	8	2.9
MSP project	5	1.8
ATE Project	3	1.1
Staff	3	1.1
Total	280	100.0

Examining the relationships between the product authors and citing work authors, different patterns of self-citation and project-program citation are revealed. Self-citations, marked in grey in the table below, are defined as when one evaluation product cites another evaluation product. In the case of ATE, there were 17 instances of self-citation, while CETP has two self-citations,

[5] Affiliation with one of the program evaluations was determined by examining the lists of project PIs and evaluators used in the Beyond Evaluation Use survey, as well as through scanning the citing works for indications of affiliation with one of the programs.

and LSC has 21 self-citations. There were no self-citations for the MSP-RETA. Also interesting is when projects cited their own and other program evaluations (marked in yellow). For example, there are three ATE projects that cited works – all of which were ATE program evaluation products. Similarly, there were 17 citations from LSC projects, all to the LSC core evaluation. The CETP projects, on the other hand, cited CETP core evaluation products 14 times and LSC core evaluation seven times. MSP projects cited LSC core evaluation products five times as well.

Table 14. Relationship between program evaluation products and citing authors

	Citation Author						
Program	**ATE Eval.**	**ATE Project**	**CETP Core**	**CETP Project**	**LSC Core**	**LSC Project**	**MSP project**
ATE	17	3	0	0	0	0	0
CETP	0	0	2	14	0	0	0
LSC	0	0	6	7	21	17	5
MSP-RETA	0	0	0	0	0	0	0
Total	17	3	8	21	21	17	5

Citing works - international origins. The citing works were predominantly US in origin, in terms of where they were published or presented. Over 93% of the works were from the US, with the rest representing the countries listed in Table 15. This may be due in some part to what languages the search engines currently catalog, however the searching was conducted with requests for all languages to be returned.

Table 15. International origins of citing works

Country of publication or presentation	**N**
Australia	1
Canada	4
China	3
Europe	2
Germany	1
Hungary	1
India	1
Mexico	1
Puerto Rico	1
Spain	1
Sweden	1
Thailand	1
UK	1
Total	19

Citing works – types of works. Citing works were categorized by type according to the same categorization scheme used with the products, with the addition of codes for "electronic resources" which was used for Web pages, archived listserv messages, or similar sources and "other." As shown in Table 16, publications (n=103) and reports (n=76) were the types of citing works most found in this analysis. Presentations (n=41) and electronic resources (n=41) were also prevalent. A smaller number of instruments or tools were found (n=7) and there was a small number of items that did not fit into these categories (n=12).

Table 16. Citing work types

Citing Work Type	N	%
Publication	103	36.8
Report	76	27.1
Presentation	41	14.6
Electronic resources	41	14.6
Other	12	4.3
Instrument/Tool	7	2.5
Total	280	100.0

Citing works – fields. The majority of the citing works came from within the STEM fields with STEM education and research related works (n=120) and STEM evaluation works (n=101) predominating. There were also a number of non-STEM education works (n=28) and non-STEM evaluation works (n=25). The six works that did not fit into the above scheme represented the fields of medicine, physics/astronomy (but not related to education), healthcare, computational biology, wellness/fitness programs, and rural development.

Table 17. Citing work fields

Citing Work Fields	N	%
STEM education/research	120	42.9
STEM evaluation	101	36.1
Education - general	28	10.0
Evaluation	25	8.9
Other	6	2.1
Total	280	100.0

Citing works – sub-fields. The citing works represented a variety of specific fields within education and evaluation. As shown in Table 18, citing works were mostly from the fields of professional development (n=76), curriculum, instrument, and assessment (n=65), and evaluation (n=54).

Table 18. Citing work sub-fields

Citing Work Sub-Fields	N	%
Professional Development	76	27.1
Curriculum/Instruction/Assessment	65	23.2
Evaluation	54	19.3
Partnerships/systemic reform	18	6.4
Workforce Development	15	5.4
Other	15	5.4
Teacher training/Pre-service	14	5.0
NSF	10	3.6
Project Capacity Building	7	2.5
Materials Development	6	2.1
Total	280	100.0

Section 3.2 Citation Patterns

This section presents findings regarding patterns of citations to the products and citation patterns across the four program evaluations. Three questions regarding product citation patterns are explored:

1. What factors are related to the number of citations a product receives?
2. What are the characteristics of highly-cited products?
3. What are the citation patterns across the four program evaluations?

3.2.1 Factors Related to Product Citation Counts

This sub-section examines variables related to the differences in the number of citations received by the 246 evaluation products. Multiple regression was used to examine the effect of the variables on predicting the number of citations for a particular product. For these analyses, the article produced by multiple evaluations [Lawrenz & Huffman, 2003] was excluded, in order to examine only the products produced by individual programs. Additionally, as only one product was coded with the field "education-general" this product was recoded as STEM education, because despite having addressed general issues in recruitment and retention, it is also applicable to STEM.

The regression model examined the relationship between a product's type, field, subfield, and program that produced the product and the number of citations that product received. This regression model is only one of many that could be used to analyze these data, but it was chosen as it reflected the interest in the main effects for each of the primary variables.

The overall regression model was found to be significant ($F=2.949$, $df=15$, $p<0.001$) although the total amount of variance in citations accounted for by the model was relatively small ($r^2=16.2\%$). Among the variables entered in the model, only the product's type was found to be related to citation levels. As discussed above, evaluation instruments and tools were cited most frequently, averaging 7.25 citations per product. This was significantly higher than the other three product types which ranged between zero citations per presentation to 1.72 citations per report.

The program evaluation producing the report and report's field and subfield were not found to be related to citation levels. Although there were significant differences between the mean number of citations of the different program evaluations in bivariate correlations, as discussed previously, program evaluation did not contribute to explaining the variance in citation counts above and beyond that explained by the product's type.

Table 19. SPSS output for regression model

Dependent Variable: Number of Citations per Product

Source	Type III Sum of Squares	df	Mean Square	F	Sig.
Corrected Model	931.395(a)	15	62.093	2.949	.000
Intercept	108.230	1	108.230	5.140	.024
Program	43.371	3	14.457	.687	.561
ProductType	450.583	3	150.194	7.133	.000
ProductField	1.375	2	.688	.033	.968
ProductFieldSpecific	19.506	7	2.787	.132	.996
Error	4821.805	229	21.056		
Total	6294.000	245			
Corrected Total	5753.200	244			

a R Squared = .162 (Adjusted R Squared = .107)

3.2.2 Characteristics of Highly-Cited Products

In order to get a more specific picture of products that were particularly influential in terms of being highly-cited, the top 5% (n=13) of cited products were chosen to be analyzed in more depth. These highly-cited works received a total of 230 of the 377 total references found in this study. In other words, the top 5% of evaluation products account for 61% of the total citations received by all 246 products.

The following table presents details about the 13 most highly-cited products:

Table 20. Highly-cited products

Product	Cita-tions	Evaluation	Date	Type	Field	Subfield
Horizon Research, Inc. (1999-2005). *Local Systemic Change through Teacher Enhancement Classroom Observation Protocol*	48	LSC	1999	Instrument/ Tool	STEM evaluation	Evaluation
Banilower, E. R., Boyd, S. E., Pasley, J. D., & Weiss, I. R. (2006). *Lessons from a decade of mathematics and science reform: A capstone report for the local systemic change through teacher enhancement initiative.* Chapel Hill, NC: Horizon Research, Inc.	33	LSC	2006	Report	STEM evaluation	Professional Development
Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. (2001-2002). *Classroom Observation Handbook.* Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	27	CETP	2001	Instrument/ Tool	STEM evaluation	Evaluation
Weiss, I. R., Montgomery, D. L., Ridgway, C. J., & Bond, S. L. (1998). *Local systemic change through teacher enhancement: Year three cross site*	25	LSC	1998	Report	STEM evaluation	Professional Development

report. Chapel Hill, NC: Horizon Research, Inc.						
Horizon Research, Inc. (1997). *Local Systemic Change 1997-98 Core Data Collection Manual.* Chapel Hill, NC: Author.	23	LSC	1997	Instrument/ Tool	STEM evaluation	Evaluation
Lawrenz, F., & Huffman, D. (2003*). How can multi-site evaluations be participatory?* American Journal of Evaluation, 24(4), 471-482.	13	Multiple	2003	Publication	Evaluation - general	Evaluation
Horizon Research, Inc. (2000-2006). *Local Systemic Change through Teacher Enhancement "Teacher Questionnaire"* = not specific as to which teacher questionnaire	11	LSC	2000	Instrument/ Tool	STEM evaluation	Evaluation
Boyd, S. E., Banilower, E. R., Pasley, J. D., & Weiss, I. R. (2003). *Progress and pitfalls: A cross-site look at local systemic change through teacher enhancement.* Chapel Hill, NC: Horizon Research, Inc.	10	LSC	2003	Report	STEM evaluation	Professional Development
Horizon Research, Inc. (2000-2006). *Local Systemic Change through Teacher Enhancement Science K-8 Teacher Questionnaire*	9	LSC	2000	Instrument/ Tool	STEM evaluation	Evaluation
Callow-Heusser, C., Chapman, H.J., & Torres, R.T. (2005). *Evidence: An Essential Tool – Planning for and Gathering Evidence using the Design-Implementation-Outcomes (DIO) Cycle of Evidence* (NSF 05-31). Prepared for the National Science Foundation by The Consortium for Building Evaluation Capacity, Utah State University.	9	MSP-RETA	2005	Report	STEM evaluation	Evaluation
Horizon Research, Inc. (2000-2006). *Local Systemic Change through Teacher Enhancement Professional Development Observation Protocol*	8	LSC	2000	Instrument/ Tool	STEM evaluation	Evaluation
Flora, D. B. & Panter, A. T. (1999). *Analysis of the psychometric structure of the LSC surveys (Technical Report).* Chapel Hill, NC: University of North Carolina at Chapel Hill.	7	LSC	1999	Report	STEM evaluation	Evaluation
Weiss, I. R., Arnold, E. E., Banilower, E. R., & Soar, E. H. (2001). *Local systemic change through teacher enhancement: Year six cross-site report.* Chapel Hill, NC: Horizon Research, Inc.	7	LSC	2001	Report	STEM evaluation	Professional Development

As shown above, the 13 most highly-cited products received between 7 and 48 citations each. Ten of the top 13 cited products were produced by the LSC core evaluation, while one was from CETP, one was from the Utah State RETA project, and one was a multiple evaluation effort. The dates of publication for the products ranged widely from 1997 to 2006. Six of the highly-

cited publications were instruments or tools, six were evaluation reports, and one was a journal article. Twelve of the thirteen most highly-cited products were related to STEM evaluation and one was a general evaluation product. In terms of specific fields, nine of the products address evaluation issues or provided evaluation tools, while the remaining four products addressed issues in professional development.

Looking more closely at the products, the majority of the works were evaluation instruments, such as classroom observation or professional development protocols, survey questionnaires, and data collection manuals. Many of the others are products that present descriptions of evaluation models or methods. The remaining highly-cited products are interim or capstone reports from the LSC evaluation, and one is a technical report regarding surveys used in the LSC core evaluation.

Characteristics of works citing the highly-cited products. Citations to the highly-cited products were split between external sources and those connected to one of the program evaluations. As shown in Table 21, approximately 55% of the citations to the highly-cited works came from sources not affiliated with the four program evaluations and their staffs, while the other 45% were from sources connected to the four program evaluations, including other program evaluations, local project evaluations, NSF, other NSF projects, or the program evaluation PIs and staff writing in other contexts.

Table 21. Authors of citing works to highly-cited products

Citing Author	N	%
Other	126	54.8
LSC Core	23	10.0
CETP Project	21	9.1
LSC Project	20	8.7
CETP Core	10	4.3
NSF Other Project	9	3.9
NSF	7	3.0
PI	7	3.0
MSP project	4	1.7
Staff	2	.9
ATE Eval	1	.4
Total	230	100.0

Although nearly half of the citations to highly-cited products came from sources connected to the program evaluations, few of the citations were self-citations. As highlighted below, only 25 of the 230 citations, or just over 10%, were citations to highly-cited products by a work produced by the same program evaluation. More interestingly, there is evidence in the table presented below that the highly-cited products were picked up to a great extent by related projects. For example, almost half of the citations to the CETP-produced highly-cited products were from CETP project sources. The LSC-related highly-cited products were cited by 20 LSC project sources, 9 CETP project sources, 8 other NSF projects, five NSF organization sources, and 4 MSP projects. This is in addition to the 98 outside sources citing the LSC highly-cited works.

Table 22. Citing work authors for each program evaluation highly-cited product

		Citation Author											Total
		ATE Eval	CETP Core	CETP Project	LSC Core	LSC Project	MSP project	NSF	NSF Other Project	Other	PI	Staff	
CETP	N	0	1	12	0	0	0	1	0	10	3	0	27
	%	.0%	3.7%	44.4%	.0%	.0%	.0%	3.7%	.0%	37.0%	11.1%	.0%	100%
LSC	N	0	9	9	23	20	4	5	8	98	3	2	181
	%	.0%	5.0%	5.0%	12.7%	11.0%	2.2%	2.8%	4.4%	54.1%	1.7%	1.1%	100%
MSP-RETA	N	0	0	0	0	0	0	0	1	8	0	0	9
	%	.0%	.0%	.0%	.0%	.0%	.0%	.0%	11.1%	88.9%	.0%	.0%	100%
Multiple	N	1	0	0	0	0	0	1	0	10	1	0	13
	%	7.7%	.0%	.0%	.0%	.0%	.0%	7.7%	.0%	76.9%	7.7%	.0%	100%
Total	N	1	10	21	23	20	4	7	9	126	7	2	230
	%	.4%	4.3%	9.1%	10.0%	8.7%	1.7%	3.0%	3.9%	54.8%	3.0%	.9%	100%

In terms of the types of citing works in which the highly-cited products were cited, the following table shows a wide variety within citing work types. As shown in Table 23, journal articles and books (n=89) and evaluation reports (n=60) were the most prominent places where the highly-cited products were referenced; although, a high number of presentations (n=35) and electronic resources (n=25) such as websites or listservs also cited the works. There is some evidence of influence of the highly-cited works on instruments and tools, as well as other types of products.

Table 23. Types of works citing the highly-cited products

Citing Work Type	N	%
Publication	89	38.7
Report	60	26.1
Presentation	35	15.2
Electronic Resource	25	10.9
Other	12	5.2
Instrument/Tool	9	3.9
Total	230	100.0

3.2.3 Citation Patterns of the Four Program Evaluations

While the data presented above convey one picture of the citations to the four evaluations' products, this sub-section presents a graphical picture of the interconnected web of the four evaluation products and citations. For the following figures, the program NetDraw was used to create maps of the networks of products and citations. In each of the maps, arrows point from the citing work to the product it is citing. The products and citing works are in the same position in each of the maps; only the colors change to highlight different characteristics of interest to describe citation patterns. This section looks at the entire network of the four evaluations, multi-evaluation product, and external citing works. Descriptions of the patterns for each individual program evaluation are presented in separate citation analysis reports written about each of the four program evaluations individually.

Networks of products and citing works. Figure 1 presents the network map for all the evaluation products and citations. In the diagram, the squares represent products produced by one of the four program evaluations or the multi-evaluation product: ATE is blue, LSC is yellow, CETP is green, MSP-RETA is purple, and the multi-evaluation product is orange. The red circles represent works authored by individuals not affiliated with any of the program evaluations.

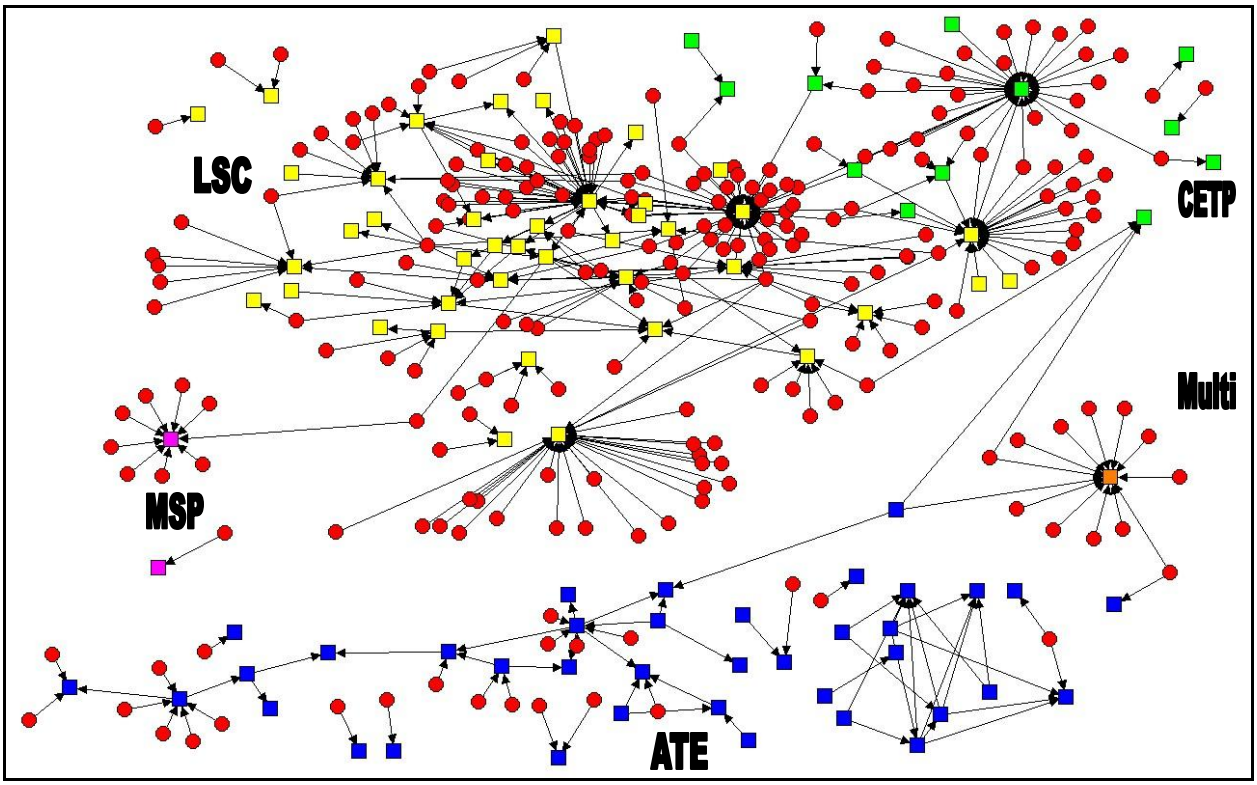


Figure 1. Citation network for the program evaluations

Color Key			
	ATE evaluation		MSP-RETA
	CETP core evaluation		Multi Evaluation
	LSC core evaluation		Other

As shown, there is an interconnected web of citations between the four program evaluations and outside sources. This web is most highly connected between and within the LSC evaluation products, which is not surprising given that the LSC products obtained the highest number of citations. When looking more closely, however, many connections between the LSC cluster and CETP cluster are apparent. There are several citations by outside sources to products produced by both evaluations, showing their joint influence in the areas of teacher preparation and professional development. The ATE products and MSP-RETA products are also connected to LSC/CETP web, although more tangentially.

Removing the external citing works helps to illustrates some of the influence the four program evaluations have had on each other. As shown in Figure 2, there are only a small number of products from one evaluation which cite another and these are primarily from the CETP evaluation to the LSC evaluation. Removing the external citations reveals that most of the citing was within program evaluations as opposed to across program evaluations.

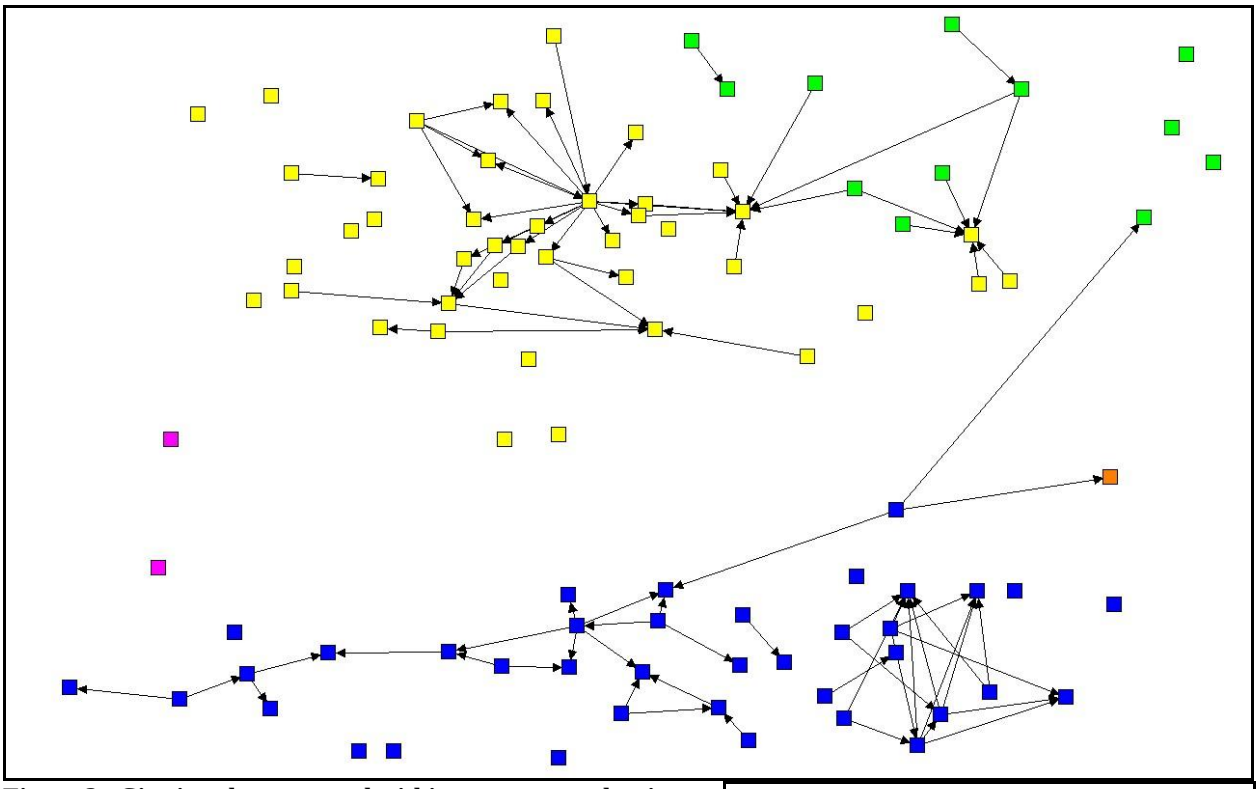


Figure 2. Citations between and within program evaluations

STEM vs. general fields. In Figure 3, the works are re-colored to represent whether the work was STEM related (in red) or related to areas outside of STEM (in blue). As shown, the majority of products were STEM-related and these were predominantly, although not exclusively, picked up by STEM-related citing works. The non-STEM related citing works are clumped around a handful of evaluation products. Two, in the bottom left-hand corner, were general evaluation-related publications produced by ATE and cited within the field of evaluation generally. One

was a report produced by the LSC evaluation, the year-three cross-site report, which was picked up in the professional development and curriculum and instruction discourse within the non-STEM education field. Finally, the multi-evaluation product, a publication about multi-site evaluation methods was picked up in the broader evaluation literature. Although there are a smattering of non-STEM works throughout the diagram, it is evident the primary influence of this body of work was within the STEM fields.

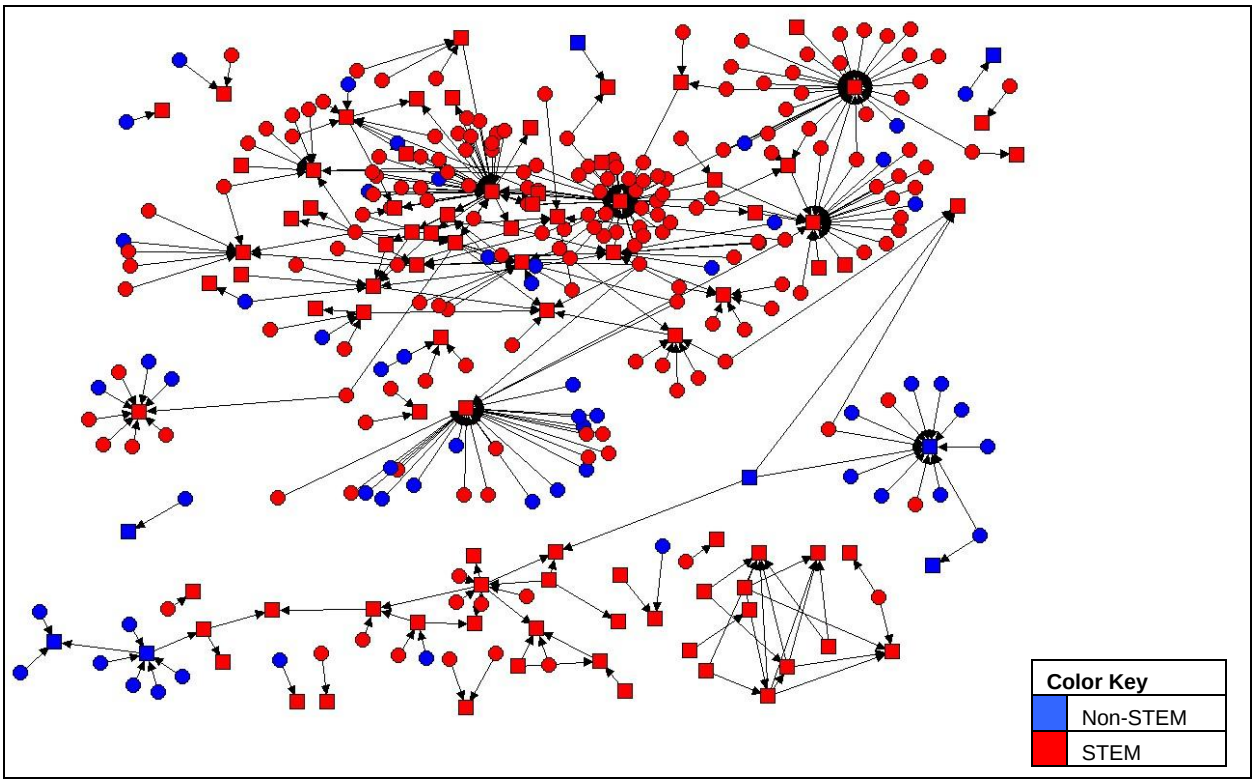


Figure 3. STEM vs. non-STEM publications

Field patterns. In Figure 4, the products and citations are all re-colored to represent the specific fields that the publication addressed; a color key is provided with the figure. The program evaluations remain represented as squares and other works are as circles. The figure shows that a wide diversity of fields is covered by the products and external works. Moreover, products are cited by a variety of fields, rather than only being captured within one area.

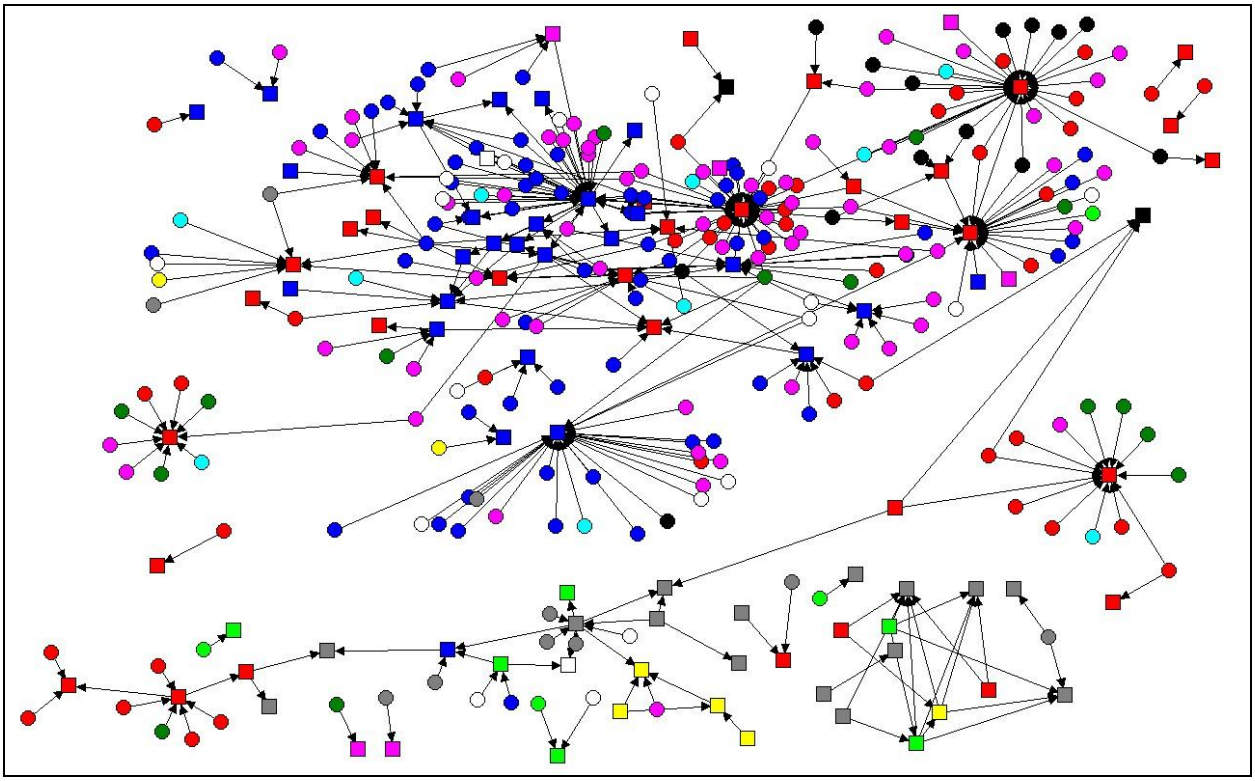


Figure 4. Fields of citing works and products

Color Key			
	Professional Development		Teacher training
	Curriculum/Instruct/Assess.		Materials Development
	Evaluation		Project Capacity Building
	Partnerships/Systemic Reform		Workforce Development
	NSF		Other

The networks of products and citations for each of the program evaluations are discussed separately in the individual citation analysis reports for each evaluation. The figures presented above present a broad picture of the four evaluations in relationship to each other and related fields.

Section 4: Discussion

The four program evaluations under consideration in this study vary in many ways, and the citation patterns of the evaluation products reflect some of these differences. As discussed, the ATE evaluation and LSC core evaluation were funded for the longest periods of time and were also specifically focused on dissemination efforts toward the end of their funding periods. Each produced the same number of products; however, the LSC products were cited more frequently than the ATE products. This difference is primarily due to the high number of citations the LSC instruments received.

While the ATE program evaluation team attempted to disseminate evaluation information to the ATE projects in novel ways, the projects were based in community colleges where publications are not a strong focus. The LSC and CETP programs, on the other hand, collaborated primarily with four-year institutions of higher education, as well as secondary schools. In both cases there are examples of LSC and CETP projects publishing their own papers in which their respective core evaluation products are cited. The CETP products and citations also show the influence of the LSC core evaluation through its citing patterns. The CETP classroom observation protocol, as well as two similar observation protocols developed by CETP projects, was based to some extent on the LSC classroom observation protocol. The citations provide evidence of this influence.

The Utah State RETA is an interesting contrast. The evaluation team provided technical assistance to local projects and created an evaluation model available for use by the local MSP projects in their evaluations. While this evaluation model was cited to some extent, there is little evidence of citations from the MSP projects to the RETA's products. The RETA was the most recently funded project, however; so, perhaps over time, citations to its products will increase.

Overall, an important conclusion to draw from this study is that evaluation instruments are being considered and used by the STEM education field. The instruments produced by each of the LSC, CETP, and MSP-RETA evaluations were among the most highly cited within this sample and within each program. After evaluation instruments, evaluation reports and articles that provide syntheses of findings and conclusions from the evaluations were also relatively highly cited. These types of summary-oriented materials provide findings that are discussed within related areas, as demonstrated by the citations to the papers within the STEM and non-STEM educational fields.

It is important to restate that this is just one snapshot in time of the citations to the evaluation products, and the sample obtained in this study, while likely representative, is an under-estimate of the actual citations in existence to the products. That said, we believe the citation patterns present one picture of the influence these program evaluations had on the STEM education and evaluation fields. The Beyond Evaluation Use project's hypothesis that there is a relationship between involvement in a program-level evaluation and subsequent use and influence of the evaluation on involved individuals appears to be supported to some extent by these data. We find evidence that related project-evaluation team members used the program evaluation products – in particular the instruments – within their local contexts, and we also see examples of high levels of citation between the many individuals involved with conducting such large scale program evaluations.

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Appendix A: Advanced Technological Education Evaluation Products

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
ATE001	Gullickson, A. R., Coryn, C. L. S., & Ritchie, L. A. (2006). Program evaluation. [aka "Project level evaluation practices"] [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #1]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Evaluation	0	
ATE002	Coryn, C. L. S., Gullickson, A. R., & Ritchie, L. A. (2006). Contributors and inhibitors influencing program improvement [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #2]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	2	ATE005, ATE004
ATE003	Ritchie, L. A., Gullickson, A. R., & Coryn, C. L. S. (2006). Workforce needs assessment [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #3]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Evaluation	0	
ATE004	Coryn, C. L. S., Gullickson, A. R., & Ritchie, L. A. (2006). Contributors and inhibitors influencing articulation agreements [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #4]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE005	Ritchie, L. A., Gullickson, A. R., & Coryn, C. L. S. (2006). Challenges and resolutions [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #5]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE006	Gullickson, A. R., Coryn, C. L. S., & Ritchie, L. A. (2006). Materials development, professional development, and program improvement: Productivity and quality [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #6]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development	2	ATE002, ATE003

ATE007	Coryn, C. L. S., Ritchie, L. A., & Gullickson, A. R. (2006). Advanced Technological Education Program Fact Sheet. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	6	ATE002, ATE006, ATE001, ATE005, ATE003, ATE004
ATE008	The Evaluation Center (2005). 2005 ATE survey fact sheet (Centers). Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE009	The Evaluation Center (2005). 2005 ATE survey fact sheet (Projects and articulation partnerships). Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE010	The Evaluation Center (2005). 2005 ATE survey fact sheet. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE011	The Evaluation Center (2004). 2004 ATE survey fact sheet. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE012	The Evaluation Center (2003). 2003 ATE survey fact sheet. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE013	Gullickson, A. R., & Hanssen, C. E. (2006). Local evaluation in multisite STEM programs: relating evaluation use and program results. In D. Huffman & F.L. Lawrenz (Eds.), New Perspectives on the Evaluation of Science, Technology, Engineering and Mathematics, New Directions in Evaluation.	Publication	STEM evaluation	Evaluation	0	
ATE014	Welch, W and Lawrenz, F. (2004). The use of evaluation in the advanced technological education program. In Evaluating the upgrading of technical courses at two year colleges: NSF's advanced technological education program. Advances in Program Evaluation Volume 9 Elsevier: London.	Publication	STEM evaluation	Evaluation	0	
ATE015	Welch, W. & Gullickson, A. (2004). The role of advisory committees. In Evaluating the upgrading of technical courses at two year colleges: NSF's advanced technological education program. Advances in Program Evaluation Volume 9 Elsevier: London.	Publication	STEM evaluation	Project Capacity Building	0	

ATE016	Gullickson, A. R., Lawrenz, F. P., & Keiser, N. (Eds.). (2004). Evaluating the upgrading of technical courses at two year colleges: NSF's advanced technological education program. Advances in program evaluation, 9. London, UK: Elsevier.	Publication	STEM evaluation	Evaluation	0	
ATE017	Gullickson, A. R., Lawrenz, F. P. & Keiser, N. (Eds.). (2002). The ATE program: Issues for consideration: A monograph. Washington, DC: National Science Foundation.	Publication	STEM evaluation	Workforce Development	1	1012
ATE018	Gullickson, A. R., & Tressler, G. R. (2002). Recruitment and retention. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 103-129). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Workforce Development	0	
ATE019	Gold, N., & Powe, K. W. (2002). Assessing the impact and effectiveness of professional development. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 81-102). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Professional Development	3	ATE029, 1169, ATE028
ATE020	Reed, L. W. (2002). Program improvement efforts: Approaches and results in improving technical education programs by ATE centers and projects. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 62-80). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Project Capacity Building	1	ATE028
ATE021	Rogers, G. R. (2002). Materials development and the ATE program. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 45-61). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Materials Development	4	ATE038, 1087, ATE039, ATE028
ATE022	Owens, T. (2002). Dissemination: A key element of the ATE program. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp.26-44). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Project Capacity Building	2	1114, 1171
ATE023	Reed, L. W. (2002). Collaboration efforts: Approaches, uses, and effectiveness of collaboration. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 10-25). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Partnerships/ systemic reform	2	ATE029, ATE028

ATE024	Lawrenz, F. P., & Keiser, N. (2002). Sustainability: Increasing the likelihood of a long-term impact by the ATE program. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 130-146). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Project Capacity Building	1	1102
ATE025	Zinser, R., & Hanssen, C. E. (2006). Improving access to the baccalaureate: Articulation Agreements and the National Science Foundation's Advanced Technological Education Program. Community College Review. Vol. 34, No. 1, 27-43	Publication	STEM evaluation	Workforce Development	0	
ATE026	Keiser, J., Lawrenz, F. P., & Appleton, J. (2004). Technical education curriculum assessment. Journal of Vocational Education Research, 29(3), 181-194.	Publication	STEM education/ research	Evaluation	2	ATE032, 1256
ATE027	Hanssen, C. E. (2004). Structural equation modeling as a tool for multisite evaluation. Dissertation Abstracts International, 65(04), 1224A. (UMI No. 3130605).	Publication	Evaluation - general	Evaluation	1	1252
ATE028	Zinser, R., & Lawrenz, F. P. (2004). New roles to meet industry needs: A look at the Advanced Technological Education program. Journal of Vocational Education Research, 29(2), 85-100.	Publication	STEM evaluation	Workforce Development	5	1181, 1232, 1250, ATE025, 1272
ATE029	Lawrenz, F. P., Keiser, N., & LaVoie, B. (2003). Sustaining innovation in technological education. Community College Review, 30(4), 1-14.	Publication	STEM education/ research	Project Capacity Building	2	1244, 1249
ATE030	Lawrenz, F. P., Keiser, N,.& LaVoie, B. (2003) Evaluative site visits: A methodological review. American Journal of Evaluation, 24(3), 341-352.	Publication	Evaluation - general	Evaluation	5	1132, 1137, 1182, 1211, 1259
ATE031	LaVoie, B. (2003). The development and validation of the biotechnology problem-solving skills assessment for community college biotechnology students. Doctoral Dissertation: University of Minnesota.	Publication	STEM evaluation	Curriculum/ Instruction/ Assessment	1	1021
ATE032	Gullickson, A. R., Wingate, L., Lawrenz, F. A., & Coryn, C. L. S. (2006). The National Science Foundation's Advanced Technology Education Program: Final evaluation report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE033	Welch, W. W., & Gullickson, A. R. (2006). The Sustainability of Advanced Technological Education Supported Efforts: An Evaluation. Kalamazoo: Western	Report	STEM evaluation	Project Capacity Building	0	

	Michigan University, The Evaluation Center.					
ATE034	Gullickson, A. R., Coryn, C. L. S., & Hanssen, C. E. (2006). ATE indicators of productivity: Six-year trends 2000-2005. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	4	ATE002, ATE006, 1174, ATE004
ATE035	Germuth, A. A., Gullickson, A. R., Lawrenz, F. P., & Hanssen, C. E. (2006). Assessing the value added by NSF's ATE program: Business and industry perspective's cross-site analysis report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	1	1174
ATE036	Coryn, C. L. S., & Hanssen, C. E. (2005). 2005 ATE technical report: Processes, procedures, and results. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	4	ATE002, ATE006, ATE001, ATE004
ATE037	Appleton, J. J., & Lawrenz, F. P. (2005). ATE materials outcome comparison: Technical report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development	0	
ATE038	Appleton, J. J., & Lawrenz, F. P. (2004). ATE materials development process report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development	0	
ATE039	Keiser, J., Lawrenz, F. P., & Appleton, J. J. (2004). Evaluation of materials produced by the ATE program. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development	2	ATE038, ATE037
ATE040	Coryn, C. L. S., Gullickson, A. R., & Hanssen, C. E. (2004). Assessing the impact and effectiveness of the Advanced Technological Education (ATE) program: 2004 survey results: Volume II status of the ATE centers. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE041	Coryn, C. L. S., Gullickson, A. R., & Hanssen, C. E. (2004). Assessing the impact and effectiveness of the Advanced Technological Education (ATE) program: 2004 survey results: Volume III status of the ATE projects. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	1	ATE025

ATE042	Jinkerson, D., & Hanssen, C. E. (2004).Evaluation of professional development efforts of ATE projects and centers: Phase I report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Professional Development	0	
ATE043	Jinkerson, D. & Hanssen, C. E. (2004). Evaluation of professional development efforts of ATE projects and centers: Phase II report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Professional Development	0	
ATE044	Jinkerson, D., & Hanssen, C. E. (2004). Evaluation of professional development efforts of ATE projects and centers: Phase III report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Professional Development	0	
ATE045	Jinkerson, D. & Hanssen, C. E. (2004). Evaluation of professional development efforts of ATE projects and centers: Phase IV report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Professional Development	0	
ATE046	Hanssen, C. E., Gullickson, A. R., & Lawrenz, F. P. (2003). Assessing the impact and effectiveness of the Advanced Technological Education (ATE) program: Survey 2003: ATE program status and trends. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	3	ATE087, ATE025, ATE028
ATE047	Lavoie, B. (2003).Summary report for the development and validation of the biotechnology problem-solving skills assessment. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment	1	1260
ATE048	Welch, W., & Reineke, R. A. (2003). National Visiting Committee (NVC) handbook. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Evaluation	0	
ATE049	Pucel, D. J. (2003). Examination of the annual survey report for 2002 and suggestions for improvement in relation to program goals, and review of survey 2002 and survey 2003 in relation to the 2002 report. New Brighton, MN: Performance Training Systems, Inc.	Report	STEM evaluation	Evaluation	0	
ATE050	Rogers, G. (2002). Developing high quality materials [brochure]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development	0	

ATE051	Lawrenz, F. P., & Keiser, N. (2002).Guide for improving sustainability [brochure]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE052	Lawrenz, F. P., Keiser, N., & Lavoie, B. (2002). A guide for planning and implementing site visits. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	Evaluation - general	Evaluation	3	ATE030, 1233, 1266
ATE053	Reineke, R. (2002). Meeting reminders for advisory committee members [brochure]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE054	Welch, W. (2002). Advisory committee checklist [brochure]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE055	National Visiting Committee (2002). Report for year 2. Daytona, FL: iTec.	Report	STEM evaluation	Workforce Development	0	
ATE056	Gullickson, A. R., & Tressler, G. R. (2002). Best practices in recruitment and retention [brochure]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	Education - general	Workforce Development	0	
ATE057	Welch, W., & Reineke, R. A. (2002). A study of advisory committees used by ATE projects and centers: Final report (Part A). Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE058	Welch, W., & Reineke, R. A. (2002). A study of advisory committees used by ATE projects and centers: Final report (Part B). Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building	0	
ATE059	Gullickson, A. R. Lawrenz, F. P., & Keiser, N. (2002). Survey 2002: The status of ATE projects and centers. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	
ATE060	Welch, W., & Gullickson, A. (2001, March). ATE evaluation project final site visit report: Advancing careers in technology and science: Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	1	ATE018
ATE061	Gullickson, A. R. Lawrenz, F. P., & Keiser, N. (2001). Survey 2001: The status of ATE projects and centers. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	0	

ATE062	Migotsky, C., & Stake, R. (2001). An evaluation of an evaluation: CIRCE's metaevaluation of the ATE program evaluation. Urbana-Champaign, IL: University of Illinois at Urbana-Champaign.	Report	STEM evaluation	Evaluation	1	ATE030
ATE063	Hartmann, D., & Loizides, G. (2001). Metaevaluation of the web-based ATE survey evaluation system. Kalamazoo, MI: Kercher Center of Social Research.	Report	STEM evaluation	Evaluation	0	
ATE064	Gullickson, A. R., Lawrenz, F. P., & Keiser, N. (2000). Status report I: The nature of the ATE program. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	1	ATE062
ATE065	Gullickson, A. R., Lawrenz, F. P., & Keiser, N. (2000). Status report II: Findings from a survey of ATE projects and centers. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development	2	ATE019, ATE062
ATE066	Site Visit Evaluation Report for the Bio Link Advanced Technology Center. Lawrenz, F., Gloria Rogers and Nanette Keiser, Report available from Bio Link Advanced Technology Center, February, 2001.	Report	STEM evaluation	Workforce Development	0	
ATE067	Site Visit Evaluation Report for the Northwest Center for Sustainable Resources. Lawrenz, F., Gloria Rogers and Leonard Sterry, Report available from the Northwest Center for Sustainable Resources, November, 2000.	Report	STEM evaluation	Workforce Development	0	
ATE068	Site Visit Evaluation Report for the Maricopa Advanced Technology Center. Lawrenz, F, Nanette Keiser and Lester Reed, Report available from Maricopa Advanced Technology Center, October, 2000.	Report	STEM evaluation	Workforce Development	0	
ATE069	Site Visit Evaluation Report for the GIS/GPS Laboratory Exercises Using Workplace Data Sets at Houston Community College (SW Campus) Project, Lawrenz, F., Nanette Keiser, Report available from Houston Community College (SW Campus), September, 2000	Report	STEM evaluation	Workforce Development	0	
ATE070	Evaluation of the NSF Advanced Technology Education Program Review Panel Meeting, Report available from National Science Foundation, December, 1999.	Report	STEM evaluation	Workforce Development	0	
ATE071	Lawrenz, F., Gullickson, A. and Barbour, S. Responses to NSF's Advanced Technology Education Program Status Survey, National Science Foundation, 1999.	Report	STEM evaluation	Workforce Development	0	

ATE072	Achieving Sustainability in ATE Projects, invited address, for the National Science Foundation at the National ATE PI meeting, Washington, DC. October, 200l	Presentation	STEM evaluation	Project Capacity Building	0	
ATE073	ATE Program Evaluation invited session Frances Lawrenz, Arlen Gullickson and Nanette Keiser for the National Science Foundation at the National ATE PI meeting, Washington, DC. October, 2001	Presentation	STEM evaluation	Evaluation	0	
ATE074	ATE Program Evaluation, Frances Lawrenz, Arlen Gullickson at NSF Advanced Technology Education national PI Conference October, 2000	Presentation	STEM evaluation	Evaluation	0	
ATE075	ATE Program Evaluation. Frances Lawrenz, Arlen Gullickson and Nanette Keiser at the NSF Advanced Technology Education Program national meeting Washington DC, October, 2002	Presentation	STEM evaluation	Evaluation	0	
ATE076	ATE Program Evaluation. Frances Lawrenz, Arlen Gullickson and Carl Hanssen at the NSF Advanced Technology Education Program national meeting Washington DC, October, 2003	Presentation	STEM evaluation	Evaluation	0	
ATE077	Coryn, C. L. S., Hanssen, C. E., Gullickson, A. R., & Ritchie, L. A. (2005). Identifying relevant evaluative criteria in the multi-site context: A conceptual approach for diverse needs, values, and goals. Paper presented at the co-sponsored American Evaluation Association/Canadian Evaluation Society conference. Toronto, Ontario, Canada. October.	Presentation	Evaluation - general	Evaluation	0	
ATE078	Demonstration session How to Conduct Site Visits. Frances Lawrenz, Nanette Keiser, at American Evaluation Association national meeting Washington, DC November, 2002	Presentation	Evaluation - general	Evaluation	0	
ATE079	Development of a Web Based Evaluation System for the Advanced Technology Education Program Frances Lawrenz, Arlen Gullickson at American Evaluation Association national meeting in Hawaii, November, 2000	Presentation	STEM evaluation	Evaluation	0	
ATE080	Dissemination of Evaluation Findings and Recommendations: Emerging Approaches. Frances Lawrenz, Arlen Gullickson and Nanette Keiser at American Evaluation Association national meeting Washington, DC November, 2002	Presentation	Evaluation - general	Evaluation	0	

ATE081	Frances Lawrenz & Arlen Gullickson, (2006). The Effect of Involvement and Evaluation Procedures on National Science Foundation Policy about Advanced Technician Education. Paper presented at the European Evaluation Society international meeting. London, England.	Presentation	STEM evaluation	Evaluation	0	
ATE082	Germuth, A., & Heinemeier, S. (2004). Defining value-added: Definitional constraints and impact on interpreting evaluation findings. Paper presented at the annual meeting of the American Evaluation Association, Atlanta, GA. November.	Presentation	Evaluation - general	Evaluation	0	
ATE083	Gullickson, A. G. (2003). Interactive video conferencing systems: An evaluation point of view. Paper presented at the annual meeting of the American Evaluation Association, Reno, NV. November.	Presentation	Evaluation - general	Evaluation	0	
ATE084	Gullickson, A. R., & Hanssen, C. E. (2004). Local evaluation in multisite programs: Does a relationship exist between evaluation use and program results? Paper presented at the annual meeting of the American Evaluation Association, Atlanta, GA. November.	Presentation	Evaluation - general	Evaluation	0	
ATE085	Hanssen, C. E. (2004). Using structural equation modeling to support formative evaluation of multisite programs. Paper presented at the annual meeting of the American Evaluation Association, Atlanta, GA. November.	Presentation	Evaluation - general	Evaluation	0	
ATE086	Hanssen, C. E., & Lawrenz, F. P. (2005). Web-based surveys in multi-site program evaluation: Strengths and limitations of two National Science Foundation cases. Paper presented at the co-sponsored American Evaluation Association/Canadian Evaluation Society conference. Toronto, Ontario, Canada. October.	Presentation	STEM evaluation	Evaluation	0	
ATE087	Hanssen, C., & Coryn, C. L. S. (2004). Theory-driven evaluations of multi-site programs. Paper presented at the annual American Evaluation Association conference. Atlanta, GA. November.	Presentation	Evaluation - general	Evaluation	0	
ATE088	How Evaluation Can Be Used to Aid in Program Sustainability: Lessons Learned Based on Quantitative and Qualitative Data. Frances Lawrenz, Arlen Gullickson and Nanette Keiser at the American Evaluation Association National Meeting St. Louis, November, 2001	Presentation	Evaluation - general	Evaluation	0	

ATE089	Issue-Oriented Papers: An Approach that Encourages Diversities of Opinions about Research and Evaluation Data for NSF's Advanced Technological Education Program. Frances Lawrenz, others in a panel at American Educational Researchers Association national meeting New Orleans, April, 2002	Presentation	STEM evaluation	Evaluation	0	
ATE090	Jinkerson, D. L., & Hanssen, C. E. (2004). Methodological issues in conducting an evaluation of professional development . Paper presented at the Annual Meeting of the Canadian Evaluation Society, Saskatoon, SA. May.	Presentation	Evaluation - general	Evaluation	0	
ATE091	Jinkerson, D. L., & Hanssen, C. E. (2005). The impact and effectiveness of the Advanced Technological Education program's professional development efforts: Using a targeted-study to identify evaluative criteria. Paper presented at the co-sponsored American Evaluation Association/Canadian Evaluation Society conference. Toronto, Ontario, Canada. October.	Presentation	STEM evaluation	Evaluation	0	
ATE092	Lawrenz, F. P. (2004). Evaluation of the quality of materials developed across multiple sites and content areas. Paper presented at the annual meeting of the American Evaluation Association, Atlanta, GA. November.	Presentation	Evaluation - general	Evaluation	0	
ATE093	Lawrenz, F. P., & Hanssen, C. E. (2003). A toolkit building workshop on site visits. Session conducted at the annual meeting of the American Evaluation Association, Reno, NV. November.	Presentation	Evaluation - general	Evaluation	0	
ATE094	Lawrenz, F. P., & Keiser, J. (2004). New directions in vocational education research. Paper presented at the annual meeting of the American Educational Research Association, San Diego, CA. April.	Presentation	STEM evaluation	Evaluation	0	
ATE095	Lessons Learned from the Development and Use of a Web-based Survey to Collect Evaluation Data. Frances Lawrenz, Arlen Gullickson and Nanette Keiser at the American Evaluation Association National Meeting St. Louis, November, 2001	Presentation	Evaluation - general	Evaluation	0	

ATE096	Schroeter, D. C., & Morell, J. (2005). Criteria for evaluating program sustainability within a multi-site context: A longitudinal approach. Paper presented at the co-sponsored American Evaluation Association/Canadian Evaluation Society conference. Toronto, Ontario, Canada. October.	Presentation	Evaluation - general	Evaluation	0	
ATE097	The Effective Use of Site Visits in the Evaluation of a National Program. Frances Lawrenz, Arlen Gullickson and Nanette Keiser at the American Evaluation Association National Meeting St. Louis, November, 2001	Presentation	Evaluation - general	Evaluation	0	
ATE098	The Effectiveness of NSF's Move into Technician Education at Community Colleges, at American Educational Research Association national meeting in New Orleans, April, 2000	Presentation	STEM education/ research	Workforce Development	0	

Appendix B: Collaboratives for Excellence in Teacher Preparation Core Evaluation Products

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
CETP001	Lawrenz, F., Gravely, A., and Ooms, A. (2006). Perceived helpfulness and amount of use of technology in science and mathematics classes at different grade levels. School Science and Mathematics, 106(3), 133-139.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment	0	
CETP002	Lawrenz, F., Huffman, D. and Appeldoorn, K. (2005). The context for learning in introductory science classes. Journal of College Science Teaching, 34(7), 40-44.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment	0	
CETP003	Lawrenz, F. & Huffman D. (2004). Using Multi-site Core Evaluation to Provide "Scientific" Evidence. Canadian Journal of Program Evaluation, 19(2),17-36.	Publication	Evaluation - general	Evaluation	1	1028
CETP004	Appeldoorn, K. (2004) Developing and validating the Collaboratives for Excellence in Teacher Preparation (CETP) core evaluation classroom observation protocol (COP). Doctoral Dissertation. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Publication	STEM evaluation	Evaluation	0	
CETP005	Lawrenz, F., Michlin, M., Appeldoorn, K., Goebel, A., & Schultz, M. (2004, March). CETP Core Evaluation: 2002-2003 Results. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Report	STEM evaluation	Teacher training/Pre-service	0	
CETP006	Lawrenz, F., Michlin, M., & Appledoorn, K., (2003, March). CETP Core Evaluation: 2001-2002 Results. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Report	STEM evaluation	Teacher training/Pre-service	3	ATE087, 1120, 1123
CETP007	Lawrenz, F., Huffman, D., Appeldoorn, K., & Michlin, M., (2002, February). CETP Core Evaluation: Field Test Results. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Report	STEM evaluation	Evaluation	2	1005, 1051

CETP008	Frantz, C., Lawrenz, F., Kushner, J., & Millar, S.B. (1999). Corridors for Collaboration. Washington, DC: National Science Foundation.	Report	STEM evaluation	Teacher training/Pre-service	2	CETP015, 1187
CETP009	Lawrenz, F., Huffman, D., and Appeldoorn, K. (2002). Classroom Observation Protocol Video Instruction Package. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM evaluation	Evaluation	1	1139
CETP010	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. (2001-2002). K-12 Surveys Handbook. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM evaluation	Evaluation	1	1141
CETP011	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. (2001-2002). Higher Education Survey Handbook. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM evaluation	Evaluation	3	1141, 1205, 1224
CETP012	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. (2001-2002). Classroom Observation Handbook. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM evaluation	Evaluation	27	1001, 1023, 1039, 1051, 1052, 1053, 1063, 1064, 1109, 1113, CETP001, 1133, 1139, 1151, 1153, 1154, 1193, 1194, 1205, 1206, 1210, 1229, 1231, 1265, 1269, 1270, 1271
CETP013	Lawrenz, F., Huffman, D., & Appeldoorn, K. (2002). Handbook for Evaluation of Activities and Assessments. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM evaluation	Evaluation	1	1147
CETP014	Lawrenz, F., & Appeldoorn. (2004, March). CETP Core Evaluation: Seeing a Glass Half Full. Presentation to the NSF Teacher Preparation PI Conference.	Presentation	STEM evaluation	Evaluation	0	
CETP015	Lawrenz, F. (2003, June). Differential Effects of Participatory Evaluation in a National Multi-site Program Evaluation. Presentation at the Canadian Evaluation Society national meeting, Vancouver, Canada.	Presentation	Evaluation - general	Evaluation	0	
CETP016	Lawrenz, F., Huffman, D., & Michlin, M. (2003, November). CETP Core Evaluation: Results Issues and the Juxtaposition of Scientific and Participatory	Presentation	STEM evaluation	Evaluation	0	

	Evaluation. Presentation at the American Evaluation Association National Meeting, Reno Nevada.					
CETP017	Lawrenz, F., & Huffman, D. (2002, November). Observing Standards-based Teaching in Science and Mathematics Classes in K-12 and Higher Education Settings. Presentation at the American Evaluation Association national meeting Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
CETP018	Lawrenz, F. (2002, April). Multicultural Aspects in the Core Evaluation of the Collaboratives for Excellence in Teacher Preparation. Pathways to Change International Conference. National Science Foundation, CETP Program, Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
CETP019	Lawrenz, F., Huffman, D., & Appeldoorn, K. (2001, November). Core Evaluation of the Collaborative for Excellence in Teacher Preparation (CETP). Presentation at the American Evaluation Association National Meeting in St. Louis.	Presentation	STEM evaluation	Evaluation	0	
CETP020	Lawrenz, F., & Appeldoorn, K. (2001, April). Workshop on the Use of the Classroom Observation Protocol five hour training session. Presentation at the Collaboratives for Excellence in Teacher Preparation national PI meeting.	Presentation	STEM evaluation	Evaluation	0	
CETP021	Lawrenz, F. (2001, April). CETP Core Evaluation Project. Plenary session panel at NSF's Collaboratives for Excellence in Teacher Preparation national PI meeting, Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
CETP022	Lawrenz, F. (2000, March). CETP Core Evaluation. Presentation at NSF national meeting, Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
CETP023	Lawrenz, F. (1998, November). Developing an evaluation core. Presentation at NSF sponsored workshop for Collaboratives for Excellence in Teacher Preparation evaluators.	Presentation	STEM evaluation	Evaluation	0	
CETP024	Lawrenz, F., Huffman, D., & Appeldorn, K. (2002). Evaluation of the Collaboratives for Excellence in Teacher Preparation (CETP). Minneapolis: University of Minnesota.	Report	STEM evaluation	Evaluation	1	1283

Appendix C: Local Systemic Change Core Evaluation Products

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
LSC001	Banilower, E. R. & Germuth A. A. (2004). Structural equation models for LSC teacher questionnaire and classroom observation data. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Evaluation	1	LSC003
LSC002	Banilower, E. R. & Shimkus, E. S. (2004). Professional development observation study. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003
LSC003	Banilower, E. R., Boyd, S. E., Pasley, J. D., & Weiss, I. R. (2006). Lessons from a decade of mathematics and science reform: A capstone report for the local systemic change through teacher enhancement initiative. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	33	1026, 1030, 1042, 1042, 1044, 1045, 1054, 1071, 1077, 1094, 1106, 1115, 1125, 1145, 1148, 1155, 1157, 1160, 1163, 1168, 1170, 1183, 1186, 1215, 1223, 1226, 1228, LSC045, 1238, LSC046, 1246, 1261, 1273
LSC004	Banilower, E. R., Rosenberg, S. L., & Weiss, I. W. (2006). Local systemic change through teacher enhancement: 2003–06 cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC005	Banilower, E.R. (2000). Local systemic change through teacher enhancement: A summary of project efforts to examine the impact of the LSC on student achievement. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	3	LSC003, 1180, LSC045
LSC006	Banilower, E.R. (2002). Results of the 2001–2002 study of the impact of the local systemic change initiative on student achievement in science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	2	LSC003, 1209

LSC007	Bowes, A. S. & Banilower, E. R. (2004). LSC classroom observation study: An analysis of data collected between 1998 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003
LSC008	Boyd, S. E., Banilower, E. R., Pasley, J. D., & Weiss, I.R. (2004). Local systemic change initiative capstone report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC009	Boyd, S. E., Banilower, E. R., Pasley, J. D., & Weiss, I. R. (2003). Progress and pitfalls: A cross-site look at local systemic change through teacher enhancement. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	10	1002, 1003, 1022, 1042, 1048, 1095, 1209, 1240, 1245, 1253
LSC010	Crawford, R. A. & Banilower, E. R. (2004). LSC Principal Questionnaire Study: A cross-sectional analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003
LSC011	Flora, D. B. & Panter, A. T. (1999). Analysis of the psychometric structure of the LSC surveys (Technical Report). Chapel Hill, NC: University of North Carolina at Chapel Hill.	Report	STEM evaluation	Evaluation	7	LSC043, LSC043, LSC014, 1110, 1217, LSC031, LSC033
LSC012	Germuth, A. A. & Banilower, E. R. (2004). Results of the 2002–03 study of the impact of the local systemic change initiative on student achievement in science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003
LSC013	Germuth, A. A., Banilower, E. B., & Shimkus, E. S. (2003). Test-retest reliability of the local systemic change teacher questionnaire. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Evaluation	1	LSC043
LSC014	Heck, D. J. & Crawford, R. A. (2004). LSC teacher questionnaire study: A longitudinal analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	2	LSC003, 1029
LSC015	Heck, D. J. & Crawford, R. A. (2004). LSC teacher questionnaire study: Indicators of systemic change, a longitudinal analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Partnerships/ systemic reform	2	LSC003, LSC045
LSC016	Heck, D. J., Rosenberg, S. L., & Crawford, R. A. (2006). LSC teacher questionnaire study: A longitudinal analysis of data collected between 1997 and 2006. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	

LSC017	Horizon Research, Inc. & Westat, Inc. (2001). Revised handbook for studying the effects of the LSC on students. Rockville, MD: Westat, Inc.	Report	STEM evaluation	Evaluation	1	1152
LSC018	Horizon Research, Inc. (1996). Local systemic change through teacher enhancement: 1995 cross-site technical report. Chapel Hill, NC: Author.	Report	STEM evaluation	Professional Development	0	
LSC019	Horizon Research, Inc. (2001). What have we learned? Local systemic change initiatives share lessons from the field. Chapel Hill, NC: Author.	Report	STEM evaluation	Professional Development	2	1116, 1165
LSC020	Horizon Research, Inc. (2005). Designing large scale education initiatives: Lessons from the local systemic change through teacher enhancement program. Chapel Hill, NC: Author.	Report	STEM education/ research	Professional Development	0	
LSC021	Horizon Research, Inc. (2006) Local systemic change through teacher enhancement: Foundations monograph. Chapel Hill, NC: Author.	Report	STEM evaluation	Professional Development	0	
LSC022	Pasley, J. (2002). The role of instructional materials in professional development: Lessons learned from the LSC community. From http://www.horizon-research.com/LSC/news/ pasley2002.php.	Report	STEM education/ research	Professional Development	3	LSC003, 1209, 1253
LSC023	Rosenberg, S. L., Heck, D. J., & Banilower, E. R. (2005). Does content preparation moderate the impacts of professional development? A longitudinal analysis of LSC teacher questionnaire data. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM education/ research	Professional Development	0	
LSC024	Shimkus, E. S. & Banilower, E. R. (2004). LSC project ratings study: An analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC025	Shimkus, E. S. & Banilower, E. R. (2004). LSC teacher interview study: An analysis of data collected between 1999 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003
LSC026	Shimkus, E.S., & Banilower, E.R. (2004, June). Local Systemic Change: Comparison of Teachers' and Principals' Perceptions of Factors Influencing Instruction. Chapel Hill, NC: Horizon Research Inc.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment	0	
LSC027	Simpson, M. A. & Banilower, E. R. (2004). LSC principal questionnaire study: A longitudinal analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	LSC003

LSC028	Simpson, M. A. & Banilower, E. R. (2004). Results of the 2003–04 Study of the Impact of the Local Systemic Change Initiative on Student Achievement in Science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC029	Weiss, I. R. (2006). Designing large scale education initiatives: Lessons from the local systemic change through teacher enhancement program. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM education/ research	Professional Development	0	
LSC030	Weiss, I. R., Arnold, E. E., Banilower, E. R., & Soar, E. H. (2000). Local systemic change through teacher enhancement: Year five cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC031	Weiss, I. R., Arnold, E. E., Banilower, E. R., & Soar, E. H. (2001). Local systemic change through teacher enhancement: Year six cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	7	LSC006, LSC012, 1152, 1158, LSC027, LSC028, 1245
LSC032	Weiss, I. R., Banilower, E. R., & Shimkus, E. S. (2004). Local systemic change through teacher enhancement: Year nine cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	1	1222
LSC033	Weiss, I. R., Banilower, E. R., Crawford, R. A., & Overstreet, C.M. (2003). Local systemic change through teacher enhancement: Year eight cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	6	1050, 1107, 1111, 1120, 1131, 1251
LSC034	Weiss, I. R., Banilower, E. R., Overstreet, C. M., & Soar, E. H. (2002). Local systemic change through teacher enhancement: Year seven cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	6	1040, 1110, 1178, 1191, 1240, 1275
LSC035	Weiss, I. R., Gellatly, G. B., Montgomery, D. L., Ridgway, C. J., Templeton, C. D., & Whittington, D. (1999). Executive summary of the local systemic change through teacher enhancement: Year four cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	4	1080, 1081, 1134, 1156
LSC036	Weiss, I. R., Gellatly, G. B., Montgomery, D. L., Ridgway, C. J., Templeton, C. D., & Whittington, D. (1999). Local systemic change through teacher enhancement: Year four cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	2	1076, 1164

LSC037	Weiss, I. R., Montgomery, D. L., Ridgway, C. J., & Bond, S. L. (1998). Local systemic change through teacher enhancement: Highlights of the year three cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC038	Weiss, I. R., Montgomery, D. L., Ridgway, C. J., & Bond, S. L. (1998). Local systemic change through teacher enhancement: Year three cross site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	25	1006, 1019, 1020, 1024, 1037, 1038, 1056, 1066, 1072, 1073, 1105, 1110, 1135, 1138, 1162, 1173, 1177, 1190, 1198, 1202, 1207, 1234, 1254, 1264, 1276
LSC039	Weiss, I. R., Rapp, K. A., & Montgomery, D. L. (1997). Local systemic change through teacher enhancement: Highlights of the year two cross site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC040	Weiss, I. R., Rapp, K. A., & Montgomery, D. L. (1997). Local systemic change through teacher enhancement: Executive summary of the year two cross site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development	0	
LSC041	Zhang, X. & Frechtling, J. (2005) Preliminary meta-analysis for local systemic change student outcome studies. Rockville, MD: Westat	Report	STEM evaluation	Professional Development	0	
LSC042	Zhang, X. & Wang, L. (2002). National science foundation local systemic change: Evaluation of data collection and analysis for cohort 1–4 projects on student achievement. Rockville, MD: Westat.	Report	STEM evaluation	Professional Development	2	LSC003, LSC045
LSC043	Banilower, E. R., Heck, D. J., & Weiss, I. R. (2007). Can professional development make the vision of the standards a reality? The impact of NSF's local systemic change through teacher enhancement initiative. Journal of Research in Science Teaching, 44(3), 375–395.	Publication	STEM education/ research	Professional Development	3	1011, 1208, 1221
LSC044	Heck, D. J., Banilower, E. B., & Weiss, I. R. (Accepted for publication). Studying the effects of effective professional development: The case of NSF's local systemic change through teacher enhancement program. Journal for Research in Mathematics Education.	Publication	STEM education/ research	Professional Development	0	

LSC045	Weiss, I. R. & Pasley, J. D. (2006, March). Scaling up instructional improvement through teacher professional development: Insights from the local systemic change initiative. CPRE Policy Briefs, RB-44.	Publication	STEM education/ research	Professional Development	5	1030, 1043, 1106, 1166, 1267
LSC046	Weiss, I. R. & Pasley, J. D. (2007). Teaching math and science: Improving instruction through local systemic change initiatives. Phi Delta Kappan, 88(9), 669–675.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment	3	1004, 1060, 1085
LSC047	Weiss, I.R. (2005). Evaluation of the nsf local systemic change program: Implications for preparing new teachers of mathematics. Association of Mathematics Teacher Educators (AMTE) Connections Vol. 15, Issue 1.	Publication	STEM evaluation	Teacher training/Pre-service	0	
LSC048	Horizon Research, Inc. (1999-2005). Local Systemic Change through Teacher Enhancement Classroom Observation Protocol	Instrument/ Tool	STEM evaluation	Evaluation	48	1010, LSC001, LSC003, 1018, 1031, 1035, 1046, 1055, 1061, 1065, 1074, 1078, 1096, 1097, 1098, 1098, 1099, 1100, 1101, 1103, 1106, 1110, 1125, CETP013, CETP012, CETP007, 1131, 1136, 1140, 1141, 1175, 1187, 1188, 1197, 1199, 1200, 1214, 1216, 1218, 1219, 1225, 1227, 1241, LSC066, 1261, LSC007, LSC009, 1268
LSC049	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Mathematics 6-12 Teacher Questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	4	1018, 1029, 1131, 1196
LSC050	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement "Teacher Questionnaire" = not specific as to which teacher questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	11	1032, 1047, 1049, 1073, LSC014, 1110, 1136, 1146, 1159, 1180, 1255 ,
LSC051	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Mathematics K-8 Teacher Questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	1	1029

LSC052	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Principal Questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	6	1029, 1032, 1138, 1159, 1200, 1248
LSC053	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Professional Development Observation Protocol	Instrument/ Tool	STEM evaluation	Evaluation	8	1029, 1032, 1041, 1161, 1172, 1176, 1220, 1278
LSC054	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Science 6-12 Teacher Questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	1	1029
LSC055	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Science K-8 Teacher Questionnaire	Instrument/ Tool	STEM evaluation	Evaluation	9	1029, 1036, 1065, 1119, 1176, 1195, 1199, LSC065, 1258
LSC056	Horizon Research, Inc. (2000-2005). Local Systemic Change through Teacher Enhancement Teacher Interview Protocol	Instrument/ Tool	STEM evaluation	Evaluation	0	
LSC057	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Teacher Interview Protocol: Final Year	Instrument/ Tool	STEM evaluation	Evaluation	0	
LSC058	Horizon Research, Inc. (2000-2005). Local Systemic Change through Teacher Enhancement Teacher Interview Summary	Instrument/ Tool	STEM evaluation	Evaluation	0	
LSC059	Horizon Research, Inc. (2000-2006). Local Systemic Change through Teacher Enhancement Teacher Interview Summary: Final Year	Instrument/ Tool	STEM evaluation	Evaluation	0	
LSC060	Horizon Research, Inc. (1997). Local Systemic Change 1997-98 Core Data Collection Manual. Chapel Hill, NC: Author.	Instrument/ Tool	STEM evaluation	Evaluation	23	1033, LSC096, 1088, 1089, 1090, 1091, 1092, 1103, 1104, 1112, 1118, CETP013, CETP012, CETP011, 1138, 1150, 1154, 1192, LSC026, 1230, 1231, 1274, CETP010
LSC061	Briars, Diane & Kaufman, Mark. (2005). Professional Development Strategies for Mathematics Improvement. Paper presented at the NSDC Annual Conference.	Presentation	STEM education/ research	Professional Development	0	
LSC062	Weiss, Iris R. (2007, Feb.) How Do You Know that You are Making Progress? Evaluating Your Professional Development Program. Paper presented at the 2007 Leadership Seminar on Mathematics Professional	Presentation	Evaluation - general	Professional Development	0	

	Development, Portland OR, February14-17, 2007.					
LSC063	Weiss, Iris R. (2006, April). Issues in Designing Professional Development for Teachers of Mathematics. Paper presented at the 2006 NCSM Meeting.	Presentation	STEM education/ research	Professional Development	0	
LSC064	Smith, P. Sean. (2005, June). Effective Professional Development in Science: Why Do We Need it? What Does It Look Like? Presentation at the National LASER K-8 Science Education Strategic Planning Institute	Presentation	STEM education/ research	Professional Development	0	
LSC065	Weiss, Iris R. (2005, May). Findings from the LSC. Presentation at the CCSSO Conference.	Presentation	STEM evaluation	Professional Development	0	
LSC066	Weiss, Iris R. (2005, November). High Quality Mathematics and Science Instruction: Where do we stand? How do we move forward? Presentation to Pittsburgh Public Schools.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment	0	
LSC067	Weiss, Iris R. (2005, April). LSCs, MSPs, SSIs, etc.: Implications of Research on Federally-Funded Projects for Improvement of Mathematics Education District-Wide. Presentation at the 2005 NCSM Meeting.	Presentation	STEM evaluation	Partnerships/ systemic reform	0	
LSC068	Weiss, Iris R. (2005, October). LSCs, MSPs, SSIs, etc.: Implications of Research on Federally-Funded Projects for Improvement of Mathematics Education District-Wide. Presentation to the North Carolina Supervisors.	Presentation	STEM evaluation	Partnerships/ systemic reform	0	
LSC069	Weiss, Iris R. (2004, November). Linking Science and Literacy in the Classroom: The National Context. Presentation at the NSTA Science and Literacy Conference.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment	0	
LSC070	Weiss, Iris R. (2005, November). Research to Practice: Applying What We Know To Improve Mathematics and Science Education. Presentation to the Network Connections Conference.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment	0	
LSC071	LSC Core Update Newsletter, September 2004	Publication	STEM evaluation	Professional Development	0	
LSC072	LSC Core Update Newsletter, June 2004	Publication	STEM evaluation	Professional Development	0	
LSC073	LSC Core Update Newsletter, March 2004	Publication	STEM evaluation	Professional Development	0	

LSC074	LSC Core Update Newsletter, December 2003	Publication	STEM evaluation	Professional Development	0	
LSC075	LSC Core Update Newsletter, September 2003	Publication	STEM evaluation	Professional Development	0	
LSC076	LSC Core Update Newsletter, June 2003	Publication	STEM evaluation	Professional Development	0	
LSC077	LSC Core Update Newsletter, March 2003	Publication	STEM evaluation	Professional Development	0	
LSC078	LSC Core Update Newsletter, December 2002	Publication	STEM evaluation	Professional Development	0	
LSC079	LSC Core Update Newsletter, September 2002	Publication	STEM evaluation	Professional Development	0	
LSC080	LSC Core Update Newsletter, June 2002	Publication	STEM evaluation	Professional Development	0	
LSC081	LSC Core Update Newsletter, March 2002	Publication	STEM evaluation	Professional Development	0	
LSC082	LSC Core Update Newsletter, December 2001	Publication	STEM evaluation	Professional Development	0	
LSC083	LSC Core Update Newsletter, September 2001	Publication	STEM evaluation	Professional Development	0	
LSC084	LSC Core Update Newsletter, July 2001	Publication	STEM evaluation	Professional Development	0	
LSC085	LSC Core Update Newsletter, March 2001	Publication	STEM evaluation	Professional Development	0	
LSC086	LSC Core Update Newsletter, September 2000	Publication	STEM evaluation	Professional Development	0	
LSC087	LSC Core Update Newsletter, July 2000	Publication	STEM evaluation	Professional Development	0	
LSC088	LSC Core Update Newsletter, March 2000	Publication	STEM evaluation	Professional Development	0	
LSC089	LSC Core Update Newsletter, April 1999	Publication	STEM evaluation	Professional Development	0	
LSC090	LSC Core Update Newsletter, September 1999	Publication	STEM evaluation	Professional Development	0	
LSC091	LSC Core Update Newsletter, November 1999	Publication	STEM evaluation	Professional Development	0	
LSC092	LSC Core Update Newsletter, May 1998	Publication	STEM evaluation	Professional Development	0	
LSC093	LSC Core Update Newsletter, August 1998	Publication	STEM	Professional	0	

			evaluation	Development		
LSC094	LSC Core Update Newsletter, November 1998	Publication	STEM evaluation	Professional Development	0	
LSC095	LSC Core Update Newsletter, January 1997	Publication	STEM evaluation	Professional Development	0	
LSC096	LSC Core Update Newsletter, March 1997	Publication	STEM evaluation	Professional Development	0	
LSC097	LSC Core Update Newsletter, May 1997	Publication	STEM evaluation	Professional Development	0	
LSC098	LSC Core Update Newsletter, November 1996	Publication	STEM evaluation	Professional Development	0	

Appendix D: Math Science Partnership – Research, Evaluation, and Technical Assistance Project Products

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
MSP001	Julnes, Patria de Lancer. (2003). Accountability assessment and the math and science partnership program. PA Times, 26(7), 12.	Publication	STEM evaluation	Partnerships/ systemic reform	0	
MSP002	Callow-Heusser, C., Chapman, H.J., & Torres, R.T. (2005). Evidence: An Essential Tool – Planning for and Gathering Evidence using the Design-Implementation-Outcomes (DIO) Cycle of Evidence (NSF 05-31). Prepared for the National Science Foundation by The Consortium for Building Evaluation Capacity, Utah State University.	Report	STEM evaluation	Evaluation	9	1007, 1034, 1057, 1082, 1093, 1212, 1213, 1223, 1282
MSP003	Callow-Heusser, C.A., Dorward, J., Lehman, S., and Chapman, H. (2007). Final Report: NSF MSP-RETA Evaluation Capacity Building Project, NSF Award Number 0233382. Washington, DC: National Science Foundation.	Report	STEM evaluation	Partnerships/ systemic reform	0	
MSP004	Worthen, B., Callow-Heusser, C.A., Dorward, J., and Lehman, S. (2004). Annual Report: NSF MSP-RETA Evaluation Capacity Building Project, NSF Award Number 0233382. Washington, DC: National Science Foundation.	Report	STEM evaluation	Partnerships/ systemic reform	0	
MSP005	Worthen, B., Callow-Heusser, C.A., Dorward, J., and Lehman, S. (2005). Annual Report: NSF MSP-RETA Evaluation Capacity Building Project, NSF Award Number 0233382. Washington, DC: National Science Foundation.	Report	STEM evaluation	Partnerships/ systemic reform	0	
MSP006	Dorward, J., Chapman, H. J., & Callow-Heusser, C. A. (2005, October). An Online Logic Model Tool and the Design-Implementation-Outcomes Cycle of Evidence: Improving Evaluation Efforts in the Math and Science Partnership Program. Paper presented at the American Evaluation Association Annual Conference, Toronto, Ontario, Canada.	Presentation	STEM evaluation	Evaluation	0	

MSP007	Chapman, H. J., Callow-Heusser, C. A., & Dorward, J. (2005, September). Establishing Partnerships to Provide Evaluation Technical Assistance and Promote Evidence-Based Designs. Paper presented at the NSF's MSP Evaluation Summit, Minneapolis, MN.	Presentation	STEM evaluation	Partnerships/ systemic reform	0	
MSP008	Callow-Heusser, C. A. (2005, January). Using evidence in Math-Science Partnerships: The DIO Cycle of Evidence. Paper presented at the National Science Foundation's MSP Learning Network Conference, Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
MSP009	Callow-Heusser, C. A., Dorward, J., Webb, N., Blank, R. (2004, November). If we build it, will they come? Insights from three NSF RETA projects funded to provide evaluation assistance to MSP projects. Panel session presented at the American Evaluation Association annual conference, Atlanta, GA.	Presentation	STEM evaluation	Evaluation	0	
MSP010	Callow-Heusser, C.A., & Dorward, J. (2004, November). Providing Evaluation Technical Assistance through Nationally-Known Evaluation Experts. Presentation at the American Evaluation Association Conference.	Presentation	Evaluation - general	Evaluation	0	
MSP011	Blank, R. (2004, November). Assisting MSPs with Empirical Data to Evaluate Professional Development and Quality of Instruction. Presentation at the American Evaluation Association Conference.	Presentation	STEM evaluation	Evaluation	0	
MSP012	Webb, N. (2004, November). Adding Value to the MSP Evaluations Project. Presentation to the American Evaluation Association Conference.	Presentation	STEM evaluation	Evaluation	0	
MSP013	Callow-Heusser, C.A., & Dorward, J. (2005, November). USU's Network for Evaluation Technical Assistance: Providing Evaluation Technical Assistance to NSF's Math-Science Partnerships. Presentation at the American Evaluation Association Conference.	Presentation	STEM evaluation	Evaluation	0	
MSP014	Shapiro, N., & Frechtling, J. (2005, November). Learning Together About Doing and Evaluating K-16 Reform Efforts. Presentation at the American Evaluation Association conference.	Presentation	STEM evaluation	Evaluation	0	
MSP015	Bocian, B., & Torres, R. (2005, November). Key Features of Collaboration between Internal and External Evaluators. Presentation at the American Evaluation Association conference.	Presentation	Evaluation - general	Evaluation	0	

MSP016	Quinn, R., & Lawrenz, F. (2005, November). Making the Unwieldy Manageable: Focusing the Evaluation of a Multi-faceted Initiative. Presentation at the American Evaluation Association conference.	Presentation	Evaluation - general	Evaluation	0	
MSP017	Callow-Heusser, CA. (October 2004). Introduction to the Evidence Framework: What evidence do you look for? Presentation at the MSP Program Meeting, Evidence: an Essential Tool, Washington, DC, Oct. 21-22, 2004	Presentation	STEM evaluation	Evaluation	0	
MSP018	Callow-Heusser, CA. (October 2004). Synergistic Evidence Gathering and Use for Multiple Evaluation Purposes. Presentation at the MSP Program Meeting, Evidence: an Essential Tool, Washington, DC, Oct. 21-22, 2004	Presentation	STEM evaluation	Evaluation	0	
MSP019	Callow-Heusser, C. A., and Schneider, S. H. (2004, January). Developing logic models for MSP projects. Workshop presented at the National Science Foundation's Math and Science Partnership Learning Network Conference, Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
MSP020	Worthen, B, Dorward, J, Lehman, S & Callow-Heusser, C. (January 2004). Evaluation Capacity Building for STEM/MSP Projects: A 3-Year Research, Evaluation, and Technical Assistance (RETA) Project. Slide presentation from the MSP PI Conference, Washington, D.C., January 2004.	Presentation	STEM evaluation	Evaluation	0	
MSP021	Callow-Heusser, C. A. (2003, November). Developing implementation evaluation models to provide assistance to the National Science Foundation's Math Science Partnerships. Paper presented at the American Evaluation Association's annual conference, Reno, NV.	Presentation	STEM evaluation	Evaluation	0	
MSP022	Dorward, J. (2003, November). Overview of the Challenges in Evaluating Math and Science Partnerships. Presentation to the American Evaluation Association conference.	Presentation	STEM evaluation	Evaluation	0	
MSP023	Callow-Heusser, C. A., & Engle, M. (2003, October). MSP partnerships: Defining and measuring partnerships and collaborations. Workshop presented at the National Science Foundation's workshop, Building Evaluation Capacity: Helping YOU Determine if Your MSP is "Working", Baltimore, MD.	Presentation	STEM evaluation	Partnerships/ systemic reform	0	

MSP024	Callow-Heusser, C. A. (2006). Evaluation Resources Database. Logan, UT: Utah State University. [available at http://www.usu.edu/cbec]	Instrument/ Tool	Evaluation - general	Evaluation	0	
MSP025	Lehman, S. (2006). Logic Model Authoring Tool. Logan, UT: Utah State University. [available at http://www.usu.edu/cbec]	Instrument/ Tool	Evaluation - general	Evaluation	1	1167

Appendix E: Multiple Evaluation Product

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
Multi001	Lawrenz, F., & Huffman, D. (2003). How can multi-site evaluations be participatory? American Journal of Evaluation, 24(4), 471-482.	Publication	Evaluation - general	Evaluation	13	1062, ATE087, 1083, 1084, 1123, 1142, 1143, 1144, 1179, 1247, 1252, 1280, 1281

Appendix F: Coding Guidelines

Variable	Variable Values	Description	Coding Procedures/Rules
Evaluation Product Reference Information (Product)	String	Reference information for evaluation product (APA format)	NA
Program Evaluation (Program)	1=ATE 2=CETP 3=LSC 4=MSP-RETA 5=Multiple	Name of program with which evaluation product is associated	NA
Evaluation Product ID Code (ProductID)	ABC(D)1234	ID code for each unique evaluation product	NA
Evaluation product type - raw categories (ProductTypeRaw)	Article Book Book chapter Briefing Paper Dissertation Electronic tool Fact Sheet Instrument Journal article Monograph Newsletter Presentation Report	Original coding scheme of evaluation product types	Provided by evaluation PIs and/or assigned by Lija
Evaluation product type – recoded (ProductType)	1=Instrument/Tool 2=Presentation 3=Publication	Instrument/Tool = Instruments, instrument manuals, electronic tools Presentation = Presentation	Recoded based on raw product type categories

	4=Report	Publication = Article, book, book chapter, monograph, dissertation, or newsletter Report = Report, briefing paper, fact sheet	
Evaluation Product Field (ProductField)	1=Education – general 2=Evaluation – general 3=STEM education/research 4=STEM Evaluation 5=Other	Education-general=education areas/issues outside STEM Evaluation-general=evaluation areas/issues outside STEM STEM education/research=general topics or research related to STEM education STEM Evaluation=topics or findings connected to a specific STEM evaluation Other=area outside the above	**Decision rules:** *STEM vs. General:* Code "STEM education/research" or "STEM evaluation" if there is a specific reference in the product's title to a STEM field, is related to a STEM program or program evaluation, or is published in a STEM-related journal. Code "Education-general" or "Evaluation-general" for products not specifically related to the STEM fields, projects, or evaluations. *Education vs. Evaluation:* Code "evaluation" (either general or STEM) if the work's title includes the name of a specific program evaluation (e.g. ATE, LSC, etc.) or is about an evaluation-related topic. Use "education" (either general or STEM) for "research-oriented" works that address general issues in education or theoretical works related to education-topics. In some cases the work may come out of a specific evaluation – however the primary focus must be on the generalizable aspect of the evaluation findings to be coded as "education" as opposed to

			"evaluation."
Evaluation Product Field – specific field (ProductFieldSpecific)	1=Teacher training/Pre-service 2=Professional development 3=Workforce development 4=Partnerships/systemic reform 5=Curriculum/Instruction/ Assessment 6=Evaluation	Teacher training/Pre-service= Teacher training and preparation pre-service including the effects of training on student achievement, etc. Professional development= Teacher training during service including the effects of PD on student achievement, etc. Workforce dev= High tech, STEM workforce training/development Partnerships/systemic reform= university, school, community partnerships and systemic reform initiatives Curriculum/instruction/assessment= related to classroom teaching: (i.e. instruction methods, curriculum, pedagogy, student assessment, standards-based instruction, math/science reform) Evaluation=evaluation-related topics such as multi-site	**Decision Rules:** Select based on key terms in title and/or abstract or full article if necessary. If the work appears to relate substantially to more than one topic, code as "other" and then specify which topics in the "Product Other Details" variable. Program evaluation reports should be coded as to the program's field, not coded as evaluation (i.e. an evaluation report about a teacher professional development program would be coded as "professional development.") The code "Evaluation" should be used only for those works addressing evaluation methods, theories, topics, and instruments/manuals. If the work is about "reform" but not specifically systemic reform or partnerships, code as "curriculum/instruction/assessment." If a work is about the effects of something (PD, training, instruction) on student achievement, code as PD, training, instruction, etc. If an evaluation report is related to a

	7=Materials development 8= Project capacity building 9= Other	evaluation theory, instrument protocols/manuals, other general evaluation theory or knowledge Materials development= development of instructional materials Project capacity building= issues related to program improvement, sustainability, advisory committee development, dissemination, replication Other= other fields	program evaluation in general code as follows: ATE = workforce development CETP = teacher training LSC = professional development MSP = partnerships
Citation reference information (Citation)	String	Citation reference information (APA format)	NA
Citation ID Code (CiteID)	1234	Numeric ID code for each unique citing reference	NA
Citation types – raw categories (CiteTypeRaw)	ATE briefing paper Book Book chapter Course syllabus Dissertation/thesis Electronic resource Grant proposal Instrument handbook Journal article Listserv message Newsletter NSF announcement Presentation Report	Original coding scheme of citation types	Assessed based on title of the publication source of work and/or other relevant details.

	RFP		
Citation Type (CiteType)	1=Instrument/Tool 2=Presentation 3=Publication 4=Report 5=Electronic resource 6=Other	Instrument/Tool = Instruments, instrument manuals, electronic tools Presentation = Presentation Publication = Article, book, book chapter, monograph, dissertation, or newsletter Report = Report, briefing paper, fact sheet Electronic resource=Website resource or listserv message Other=other	Recoded from above categories
Citation Field (CiteField)	1= Education – general 2= Evaluation – general 3= STEM education/research 4= STEM Evaluation	Education-general= education areas/issues outside STEM Evaluation-general= evaluation areas/issues outside STEM STEM education/research= general topics or research related to STEM education STEM Evaluation= topics or findings connected to a specific STEM evaluation	**Decision rules:** *STEM vs. General:* Code "STEM education/research" or "STEM evaluation" if there is a specific reference in the work's title to a STEM field, is related to a STEM program or program evaluation, or is published in a STEM-related journal. Code "Education-general" or "Evaluation-general" for works not specifically related to the STEM fields, projects, or evaluations. *Education vs. Evaluation:* Code

	5= Other	Other= area outside the above	"evaluation" (either general or STEM) if the work's title includes the name of a specific evaluation or is about an evaluation-related topic. Use "education" (either general or STEM) for "research-oriented" works that address general issues in education or theoretical works related to education-topics. In some cases the work may come out of a specific evaluation – however the primary focus must be on the generalizable aspect of the evaluation findings to be coded as "education" as opposed to "evaluation."
Citation Field – specific field (CiteFieldSpecific)	1=Teacher training/Pre-service 2=Professional development 3=Workforce development 4=Partnerships/systemic reform	Teacher training/Pre-service= Teacher training and preparation pre-service including the effects of training on student achievement, etc. Professional development= Teacher training during service including the effects of PD on student achievement, etc. Workforce dev= High tech, STEM workforce training/development Partnerships/systemic reform= university, school, community partnerships and systemic reform initiatives	**Decision Rules:** Select based on key terms in title and/or abstract. If the work appears to relate substantially to more than one topic, code as "other" and then specify which topics in the "Product Other Details" variable. Program evaluation reports should be coded as to the program's field, not coded as evaluation (i.e. an evaluation report about a teacher professional development program would be coded as "professional development.") The code "Evaluation" should be used only for those works addressing evaluation methods, theories, and topics.

	5=Curriculum/Instruction/ Assessment	Curriculum/instruction/assessment= related to classroom teaching: (i.e. instruction methods, curriculum, pedagogy, student assessment, standards-based instruction, math/science reform)	If the work is about "reform" but not specifically systemic reform or partnerships, code as "curriculum/instruction/assessment." If a work is about the effects of something (PD, training, instruction) on student achievement, code as PD, training, instruction, etc.
	6=Evaluation	Evaluation= evaluation-related topics such as multi-site evaluation theory, instrument protocols/manuals, other general evaluation knowledge	If an evaluation report is related to a program evaluation in general code as follows:
	7=Materials development	Materials development= development of instructional materials	ATE = workforce development CETP = teacher training LSC = professional development MSP = partnerships
	8= Project capacity building	Project capacity building= issues related to program improvement, sustainability, advisory committee development, dissemination, replication	
	9= Other	Other= other fields	
	10=NSF	NSF=NSF related materials, including RFPs, COVs, program solicitations, press releases	
Citation Author – raw categories (CiteAuthorRaw)	??? ATE Centers Evaluation ATE Program Evaluation	Original coding categories for citation author relationship to program evaluations	NA

	ATE project CETP Core Evaluation CETP project inter-ATE program eval source inter-LSC program eval source LSC project representative MSP-RETA eval team MSP project NSF - other project NSF NSF insider outsider self-citation		
Citing Author (CiteAuthor)	ATE Eval ATE Project CETP Core CETP Project LSC Core LSC Project MSP project NSF NSF Other Project Other	Authors listed as related to an evaluation or project if the work is produced as part of said evaluation or project. NSF = official NSF documents. NSF Other Project = work produced by an NSF supported project, not one of four programs of interest Other=doesn't fit into any other category	Author identities determined by whether they are listed as one of the authors on one of the four evaluation's product lists (then they are related to that program evaluation) or if they are on the list of project evaluators and PIs used for the survey (then they are project "staff").

	PI Staff	PI=work authored or co-authored by an evaluation PI but not in the context of the four evaluations Staff=work authored by a program eval staff person but not in the context of the four evaluations	
Citing Author – Specific Details (CiteAuthorSpecific)	String	Specific name of author, organization, or PI if a PI is one of the authors.	
Relationship between citing work and product (Relationship)	1=Self-citation 2=Within program citation 3=Across program citation 4=Across project-program citation	Self-citation=at least one person is listed as both an author of the product and an author of the citing work Within program citation=evaluation product citing another evaluation product within the same program evaluation (but none of the author names are the same) Across program citation=evaluation product from one evaluation citing an evaluation product from one of the other program evaluations Across project-program citation=work produced by an ATE/CETP/LSC/MSP project	Relationships determined by the citing author variable above and the program evaluation name. Can be computed based on the two variables and cross-checked against the manual coding Lija did. Issues: Frances Lawrenz and the other PIs may have played multiple roles – e.g. she was the CETP program evaluator, an ATE program evaluation contractor, and LSC project evaluator, and also wrote works not affiliated with any of these evaluations. Care should be taken to identify what her specific role was for any given work, and then determine the relationship between that role and the evaluation product it is citing.

	6=Within project-program citation 7=Other 8=Unrelated to program evaluation	citing an evaluation product from a different program evaluation Within project-program citation=work produced by an ATE/CETP/LSC/MSP project citing an evaluation product from the same program evaluation Doesn't fit into the above – some relationship but not as above (e.g. work produced by Frances but not as the CETP evaluator) – Unrelated to program evaluation=Work is not related to any program or project evaluation	Use the "Other" code for works produced by individuals such as Frances, other PIs, or project evaluators who have some connection to a program or project evaluation, but the specific work was not produced as part of a specific evaluation of interest.
International (International)	String	Country of publication or presentation of work if outside the US	
Google Scholar (GS)	1=yes 2=no	Google scholar was source of citation	
Web of Science (WoS)	1=yes 2=no	Web of Science was source of citation	
Google (Google)	1=yes 2=no	Google.com was source of citation	

Appendix G: Web of Science Search Terms

Evaluation	Years Searched	Search terms
ATE	1997-2007	Gullickson, A*, Coryn, C*, Ritchie, L*, Evaluation Center (and variants), Western Michigan University (and variants), Welch, W*, Lawrenz, F*, Gold, N*, Reed, L*, Rogers, G*, Owens, T*, Zinser, R*, Lavoie, B*, Hanssen, C*, Appleton, J*, Germuth, A*, Pucel, D*, Rogers, G*, Reineke, R*, Migotsky, C*, Hartmann, D*, Schroeter, D*, Morell, J*, Jinkerson, D*, Pucel, D*, Keiser, N*, Advanced Technology Education (and variants, including ATE). Also looked for all article titles using the general search function.
CETP	1998-2007	Lawrenz F*, Appledoorn K*, Huffman D*, Ooms A*, Gravely A*, Michlin M*, Goebel A*, Schultz M*, Frantz C*, Sun T*, Millar S*, Kushner J*, NSF (and variants), CETP, Collaboratives for Excellence in Teacher Preparation.
LSC	1996-2007	Weiss, IR, Banilower, ER, Germuth, AA, Shimkus, Boyd, SE, Pasley, JD, Rsenberg, SL, Bowes, AS, Crawford, RA, Flora, DB, Horizon Research (and variants including HR, HRI), Westat*, Simpson, MA, Frechtling, J*, Briars, D*, Kaufman, M*, Smith, PS
MSP	2002-2007	Julnes, P*, Callow-Heusser, C*, Callow, C*, Heusser, C*, Altschuld, J*, Torres, RT, Dorward, J*, Blank, R*, Webb, N*, Shapiro, N*, Frechtling, J*, Worther, B*, Haynie, K*, Schneider, S*, Quinn, R*, Bocian, K*, Math Science Partnership (and variants including MSP), Utah State

Note: An asterisk (*) represents a wildcard allowing for any variations following a particular letter.

Appendix H: Citing Work References and Descriptions

Citation ID	Citing Work Reference	Citation Type	Citation Field	Citation Subfield
1001	633 Study Group. (nd). Chapter 2 Developing a Research Proposal. Available from: http://663studygroup.pbwiki.com/Chapter%202%20%20Developing%20a%20Research%20Proposal	Electronic Resource	STEM education/ research	Evaluation
1002	Abell, S., Ehlert, M., Lannin, J., Marra, R., Cole, J., Park Rogers, M., Lee, M., Wang, C. & Musikul, K. (2005, April). Profiles of Mathematics and Science Teacher Professional Development Programs: Theoretical Basis, Activities, and Effectiveness. Paper presented at the annual meeting of the American Educational Research Association, Montreal, Quebec.	Presentation	STEM education/ research	Professional Development
1003	Abell, S.K, Lannin, J.K., Marra, R.M., Ehlert, M.W., & Cole, J.S. (2007). Multi-site evaluation of science and mathematics teacher professional development programs: The project profile approach. Studies in Educational Evaluation, 33(2), 135-158.	Publication	STEM evaluation	Evaluation
1004	Access My Library. (2007). Teaching math and science: Improving instruction through Local Systemic Change initiatives. Available from: http://www.accessmylibrary.com/coms2/summary_0286-31239646_ITM	Electronic Resource	STEM evaluation	Professional Development
1005	Adamson, S.L., Banks, D., Burtch, M., Cox, F., Judson, E., Turley, J.B., Benford, R., & Lawson, A.E. (2003, December). Reformed undergraduate instruction and its subsequent impact on secondary school teaching practice and student achievement. Journal of Research in Science Teaching, 40(10), 939-957.	Publication	STEM education/ research	Teacher training/ Pre-service
1006	Alabama Mathematics, Science, and Technology Initiative Committee. (2000, October). Report on the review of the literature. Presented to the Alabama State Board of Education.	Report	STEM education/ research	Partnerships/ systemic reform
1007	American Society for Biochemistry and Molecular Biology. (nd). K-12 Outreach Resources on the Web. Retrieved from http://www.faseb.org/asbmb/epd/orresources.html.	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment

1010	Arizona Collaborative for the Excellence of Preparation of Teachers. (nd). About RTOP. Available from: http://physicsed.buffalostate.edu/AZTEC/RTOP/RTOP_full/about_RTOP.html	Report	STEM evaluation	Evaluation
1011	Astrophysics Data System. (nd). Digital Library for Physics and Astronomy. Available from: http://www.adsabs.harvard.edu/	Electronic Resource	Other	Other – Physics/ astronomy
1012	Bailey, Thomas R.; Matsuzuka, Yukari; Jacobs, James; Morest, Vanessa Smith; Hughes, Katherine L. (2003). Institutionalization and Sustainability of the National Science Foundation's Advanced Technological Education Program. ERIC document: ED482183.	Report	STEM evaluation	Project Capacity Building
1018	Basista, B., & Mathews, S. (2002). Integrated Science and Mathematics Professional Development Programs. School Science and Mathematics, 102(7), 359.	Publication	STEM education/ research	Professional Development
1019	Beasley, W. (2000). Systemic Reform through Teacher Change: congruence and contradiction. Teaching Education, 11(1), 39.	Publication	Education - general	Partnerships/ systemic reform
1020	Center for Science, Mathematics, and Engineering Education. (1999). Designing Mathematics or Science Curriculum Programs: A Guide for Using Mathematics and Science Education Standards. Washington, DC: The National Academies Press.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1021	China Medical College collection number: 743. http://211.20.11.205:8080/edissstat/customPDFList?CusNO=3481&NO=743	Electronic Resource	Other	Other - Medicine
1022	Chrispeels, J. (2004). Learning to lead together: The promise and challenge of sharing leadership. Thousand Oaks, CA: Sage Publications Inc	Publication	Education - general	Other - Leadership
1023	Collaborative for Excellence in Teacher Preparation in Pennsylvania. (2002, June). CETP-PA Definitions & Terms: Definitions Related to Course Revision. Harrisburg, PA: author. Available from: http://www.cetp-pa.iup.edu/terms.htm	Report	STEM evaluation	Teacher training/ Pre-service
1024	Commission of the European Communities. (2007). Accompanying document to the proposal for a recommendation of the European Parliament and of the Council on the quality of teacher education. Brussels: author.	Report	Education - general	Teacher training/ Pre-service

1026	Connecticut Department of Higher Education. (2006, September). 2007 Teacher Quality Partnership Grant Program: Request for Proposals and Application Guidelines. Hartford, CT: author.	Other	Education - general	Partnerships/ systemic reform
1028	Coryn, C.L.S. (2005, February). Global Review: Publications: Canadian Journal of Program Evaluation, Volume 19(2), Fall 2004. Journal of MultiDisciplinary Evaluation, 2.	Publication	Evaluation	Evaluation
1029	Council of Chief State School Officers. (2006, September 9). Study of Math and Science Professional Development Programs: Improving Evaluation of Professional Development. Washington, DC: author. Available from: www.ccsso.org/content/PDFs/Eval_Tools_Chart-Change_Instruction.xls	Electronic Resource	STEM evaluation	Professional Development
1030	Council of Chief State School Officers. (2007, November). Resources for State Leaders and Evaluators: Improving Evaluation of Professional Development. Washington, DC: CCSSO. Available from http://www.ccsso.org/projects/improving_evaluation_of_professional_development/Resources/	Electronic Resource	STEM evaluation	Professional Development
1031	Czerniak, C.M. & Beltyukova, S. (2003, July). Tapestries: A successful elementary science curriculum in metropolitan Toledo. Urban Affairs Center: The University of Toledo.	Report	STEM education/ research	Curriculum/ Instruction/ Assessment
1032	Czerniak, C.M. (2003). Christa McAuliffe Award: 2003 Final Proposal Submission for the Toledo Area Partnership in Education: Support Teachers as Resources to Improve Elementary Science (TAPESTRIES). Available from: http://www.aascu.org/programs/mcauliffe/pdf/toledo.pdf	Other	STEM education/ research	Curriculum/ Instruction/ Assessment
1033	D'Ambrosio, B. S., Boone, W.J., & Harkness, S.S. (2004, January). Planning District-Wide Professional Development: Insights Gained From Teachers and Students Regarding Mathematics Teaching in a Large Urban District. School Science and Mathematics, 104(1), 5-15.	Publication	STEM education/ research	Professional Development
1034	D'Avanzo, C., et al. (2006). Design and evaluation of TIEE, a peer-reviewed electronic teaching resource. Frontiers in Ecology and the Environment, 4(4), 189-195.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment

1035	Danin, S. (2000, August 11). Re: Classroom Observation Protocol. Message posted on the Discussion: Intensive Sites Task Force for E=MSC2 listserv. Available from: http://www.edgateway.net/cs/ege/forum/cs_disc/573?x-showcontent=message_text.	Electronic Resource	STEM evaluation	Professional Development
1036	Davidson, A.B. (2000, November). Eighth Grade Science Teachers Use of Instructional Time: Comparing Questions from the Third International Mathematics and Science Study (TIMSS) and National Science Foundation Questionnaires. Paper presented at the Annual Meeting of the Mid-South Educational Research Association (Bowling Green, KY, November 15, 2000). Eric document# ED448056.	Presentation	STEM evaluation	Curriculum/ Instruction/ Assessment
1037	Desimone, L.A., Smith, T.M., & Ueno, K. (2006). Are teachers who need sustained, content-focused professional development getting it? An administrator's dilemma. Educational Administration Quarterly, 42(2), 179-215.	Publication	Education - general	Professional Development
1038	Desimone, L.M., Porter, A.C., Garet, M.S., et al. (2002). Effects of professional development on teachers' instruction: Results from a three-year longitudinal study. Educational Evaluation and Policy Analysis, 24(2), 81-122.	Publication	Education - general	Professional Development
1039	Dirr, P.J. (nd.) Classroom Observation Protocols: Potential Tools for Measuring the Impact of Technology in the Classroom. Policy and Planning Series #104. Alexandria, VA: Appalachian Technology in Education Consortium.	Report	Evaluation	Curriculum/ Instruction/ Assessment
1040	Donovan, M.S., & Pellegrino, J.W. (2004). Learning and Instruction: A SERP Research Agenda. Washington, DC: The National Academies Press.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1041	Duggan-Haas, D. Using The Secondary Teaching Analysis Matrix in Professional Development. Presentation available from: people.colgate.edu/ddugganhaas/STAM/STAM%20NSTA%20Workshop.ppt	Presentation	Education - general	Professional Development
1042	Committee on Science Learning, Kindergarten through Eighth Grade, Richard A. Duschl, Heidi A. Schweingruber, and Andrew W. Shouse, Editors. (2007). Taking Science to School: Learning and Teaching Science in Grades K-8. Washington, DC: The National Academies Press.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment

1043	Education Commission of the States Education Policy Issue Site: Mathematics. (2007). Selected Research and Readings. Available from: http://www.ecs.org/html/IssueSection.asp?issueid=81&s=Selected+Research+%26+Readings	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment
1044	Education Commission of the States. (nd). Document Number: 6937. Available from http://www.ecs.org/html/Document.asp?chouseid=6937.	Report	Education - general	Professional Development
1045	Educational Research Associates. (2006, April). Successes, Challenges Cited in Math, Science Reform. E-bulletin, 33(15).	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1046	Ellis, M.W., Malloy, C.E., Meece, J.L., & Sylvester, P.R. (2007). Convergence of observer ratings and student perceptions of reform practices in sixth-grade mathematics classrooms. Learning Environments Research, 10(1), 1-15.	Publication	STEM evaluation	Curriculum/ Instruction/ Assessment
1047	Eun, B. & Heining-Boynton, A.L. (2007, Sept.). Impact of an English-as-a-Second-Language Professional Development Program. The Journal of Educational Research, 101(1), 36-49.	Publication	Education - general	Professional Development
1048	Evaluation of Professional Development Projects. (nd). Resources. Missouri: author. Available from: http://www.pdeval.missouri.edu/resources_2.html	Electronic Resource	Evaluation	Professional Development
1049	Ferrini-Mundy, J. (2004). SIRG Brief 3: What Does It Mean to Be Standards-based? Issues in Conceptualizing, Measuring, and Studying Alignment with Standards. In Frank K. Lester, Jr., and Joan Ferrini-Mundy (Eds.) Proceedings of the National Council of Teachers of Mathematics Research Catalyst Conference.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1050	Fletcher, C., Meyer, J.D., Barufaldi, J.P., Lee, E., Tinoca, L., & Bohman, T.M. (2004, March). The Science Classroom Profile: Design, Development, and Use. Paper presented at the National Association of Research in Science Teaching conference, Vancouver, Canada.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1051	Flick, L., Morrell, P. D., & Wainwright, C. (2004). Reform Teaching Strategies Used by Student Teachers. School Science and Mathematics, 104(5), 199-213.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment

1052	Flugman, B., Schroder, B., & Gningue, S. (2004, March.). Follow-up Study of the New York Collaborative for Excellence in Teacher Preparation. Presentation to the Teacher Preparation PI Conference, NSF CETP, STEMTP, ATE Programs, Arlington, VA.	Presentation	STEM evaluation	Teacher training/ Pre-service
1053	Fouts, J.T., Brown, C., & Thieman, G.Y. (2002, September). Classroom Instruction in Gates Grantee Schools: A Baseline Report. Bothell, WA: Fouts and Associates.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
1054	Fraser, B. J. & Kahle, J. B. (2007). Classroom, Home and Peer Environment Influences on Student Outcomes in Science and Mathematics: An analysis of systemic reform data. International Journal of Science Education, 1-19.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1055	Frechtling, J. (2001, February 9). Responses to Questions from LSC Conference Participants. Paper presented at the Studying the Effects of the LSC on Students Conference.	Presentation	STEM evaluation	Professional Development
1056	Garet, M.S., Porter, A.C., Desimone, L., et al. (2001). What makes professional development effective? Results from a national sample of teachers. American Educational Research Journal, 38(4), 915-945.	Publication	Education - general	Professional Development
1057	Georgia Tech College of Computing - NSF Resource Office. (nd). NSF Solicitations and Important Announcements. Retrieved from http://www.cc.gatech.edu/program/NRO/.	Electronic Resource	STEM education/ research	NSF – RFP
1060	Goliath Business Knowledge on Demand. (2007). Teaching math and science: Improving instruction through Local Systemic Change initiatives. Available from: http://goliath.ecnext.com/coms2/gi_0199-6619640/Teaching-math-and-science-improving.html	Electronic Resource	STEM evaluation	Professional Development
1061	Good, K. (2000, September 18). Re: Classroom Observation Protocol. Message posted on the Discussion: Intensive Sites Task Force for E=MSC2 listserv. Available from: http://www.edgateway.net/cs/ege/forum/cs_disc/573?x-showcontent=message_text.	Electronic Resource	STEM evaluation	Professional Development
1062	Goodyear, L. (2006, September 20). How can program evaluation contribute to understanding how diverse learners benefit from ISL? Presentation to the Committee on Learning Science in Informal Environments. Washington, DC: The National Academies Board on Science Education.	Presentation	STEM evaluation	Curriculum/ Instruction/ Assessment

1063	Goss, M. (2003, May 1). Authentic Assessment of Pre-Service Students' Performance with a Palm or iPAC Handheld Tool. TechLearning Newsletter. Available from: http://www.techlearning.com/db_area/archives/WCE/archives/marlgoss.php	Publication	STEM evaluation	Teacher training/ Pre-service
1064	Goss, M., & Newcombe, E. (nd). Summary: NSF CETP-PA Level II Supervision Grant: Partnership for Excellence in Supervision. Available from: http://www.wcupa.edu/_ACADEMICS/CoEd/EdTech/cept_pa.htm	Report	STEM evaluation	Teacher training/ Pre-service
1065	Grable, L.L. & Park, J.C. (2002, February). Anchoring Inquiry And Content For Teachers Through Technology: Middle School Mathematics And Science Integration. In B.S. Wojnowski & D.G. Haase, (Eds). Proceedings of the Conference on K-12 Outreach from University Science Departments. Raleigh, NC: The Science House, North Carolina State University.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1066	Grant, M.L. (2005). Impact of the Kindergarten Teacher Reading Academy on the Instructional Practices of Kindergarten Teachers. Doctoral Dissertation: Mississippi State University.	Publication	Education - general	Curriculum/ Instruction/ Assessment
1071	Hake, R. (2006, March). [Phys-l] Lessons from a Decade of Mathematics and Science Reform. Electronic posting to the Phys-L: Forum for Physics Educators Listserv. Archived at: https://carnot.physics.buffalo.edu/archives/2006/03_2006/msg00005.html	Electronic Resource	STEM education/ research	Professional Development
1072	Hamilton, L.S. & Guarino, C.M. (2004, November). Working Paper: Measuring the Practices, Philosophies, and Characteristics of Kindergarten Teachers. RAND Education Working Paper Series, WR-199-EDU. Prepared for the National Center for Education Statistics by RAND Education, Santa Monica, CA.	Report	Education - general	Curriculum/ Instruction/ Assessment
1073	Hamilton, L.S., McCaffrey, D.F., Stecher, B.M., et al. (2003). Studying large-scale reforms of instructional practice: An example from mathematics and science. Educational Evaluation and Policy Analysis, 25(1), 1-29.	Publication	STEM evaluation	Partnerships/ systemic reform

1074	Haney, J.J., Lumpe, A.T., Czerniak, C.M. & Egan, V. (2002, August). From Beliefs to Actions: The Beliefs and Actions of Teachers Implementing Change. Journal of Science Teacher Education, 13(3), 171-187.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1076	Hartshorne, R. (2005). Effects of Integrating Hypermedia into Elementary Science Professional Development on Science Content Knowledge. Journal of Science Education and Technology, 14(4), 415.	Publication	STEM education/ research	Professional Development
1077	Harvard Graduate School of Education. (2006, March/April). WIDE World Email Newsletter. Available from http://learnweb.harvard.edu/wide/news/newsletter/2006_04.cfm	Publication	Education - general	Professional Development
1078	Heck, D. (2001, February). Proposal for Measuring the LSC Program's Effects on Student Achievement in Science. Paper presented at the Studying the Effects of the LSC on Students Conference.	Presentation	STEM evaluation	Professional Development
1080	Heller, J.I., Daehler, K.R., & Shinohara, M. (2003, Fall). Connecting all the Pieces: Mosaic approach to evaluation makes a complete picture. Journal of Staff Development, 24(4).	Publication	Evaluation	Evaluation
1081	Heller, J.I., Daehler, K.R., Shinohara, M., & Kaskowitz, S.R. (2004, April). Fostering Pedagogical Content Knowledge about Electric Circuits Through Case-Based Professional Development. Paper presented at the annual meeting of the National Association for Research on Science Teaching, Vancouver, Canada.	Presentation	STEM education/ research	Professional Development
1082	Heywood and Middleton Primary Care Trust. (2006). Draft Evidence-Based Action Plans, Healthy Lifestyles 2006-2010.	Report	Other	Other - Healthcare
1083	High, C., & Nemes, G. (2005, August). Evaluating rural development: Mediating social learning within LEADER. Paper for Working Group 12, XXI European Congress of Rural Sociology, Keszthely, Hungary.	Presentation	Evaluation	Other – Rural development
1084	High, C., & Nemes, G. (2007). Social learning in LEADER: Exogenous, endogenous and hybrid evaluation in rural development. Sociologia Ruralis, 47(2), 103-119.	Publication	Evaluation	Other – Rural development
1085	HighBeam Research. (2007). Teaching math and science: Improving instruction through Local Systemic Change initiatives. Available from: http://www.highbeam.com/doc/1G1-163471125.html	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment

1087	Huang, C. (2005). Designing high-quality interactive multimedia learning modules. Computerized Medical Imaging and Graphics, Volume 29, Issue 2-3, Pages 223-233.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1088	Huffman, D, & Kalnin, J. (2003, August). Collaborative inquiry to make data-based decisions in schools. Teaching and Teacher Education. 19(6):569-580	Publication	Education - general	Project Capacity Building
1089	Huffman, D. & Lawrenz, F. (2004). The Impact of a State Systemic Initiative on U.S. Science Teachers and Students. Journal International Journal of Science and Mathematics Education, 1(3), 357-377.	Publication	STEM education/ research	Partnerships/ systemic reform
1090	Huffman, D. (2006, June). Reforming Pedagogy: Inservice Teacher Education and Instructional Reform. Journal of Science Teacher Education, 17(2), 121-136.	Publication	STEM education/ research	Professional Development
1091	Huffman, D., Goldberg, F., & Michlin, M. (2003). Using computers to create constructivist learning environments: Impact on pedagogy and achievement. Chesapeake, VA: Association for the Advancement of Computing in Education. Available from: http://www.thefreelibrary.com/Using+computers+to+create+constructivist+learning+environments:...-a0111266274	Report	STEM education/ research	Curriculum/ Instruction/ Assessment
1092	Huffman, D., Thomas, K. & Lawrenz, F. (2003, December). Relationship Between Professional Development, Teachers' Instructional Practices, and the Achievement of Students in Science and Mathematics. School Science and Mathematics, 103(8), 78-87.	Publication	STEM education/ research	Professional Development
1093	International Society for Computational Biology. (2005). Government reports and recommendations. Retrieved from http://www.iscb.org/weeklyDocs/june05.shtml.	Electronic Resource	Other	Other – Computational biology
1094	Inverness Research Associates. (2006, November). The Gilbert Elementary Science Program: The Infrastructure Needed to Sustain High Quality Instruction. Inverness, CA: author.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
1095	Inverness Research Associates. (2006, June). The Biological Sciences Curriculum Study, National Academy for Curriculum Leadership: Contributions and Lessons Learned: An Evaluation Brief. Inverness, CA: author.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment

1096	Iona, S. (nd). Summary of Classroom Observation Protocols. Available from: http://www.colorado.edu/physics/phys4810/phys4810_fa06/protocol.html	Electronic Resource	STEM evaluation	Evaluation
1097	Jeanpierre, B.J. (2004). Two Urban Elementary Science Classrooms: The Interplay between Student Interactions and Classroom Management Practices. Education, 124(4), 664.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1098	Johnson, C.C. (2007, August). Whole-School Collaborative Sustained Professional Development and Science Teacher Change: Signs of Progress. Journal of Science Teacher Education, 18(4), 629-661.	Publication	STEM education/ research	Professional Development
1099	Johnson, C.C. (2007, April). Technical, political and cultural barriers to science education reform. International Journal of Leadership in Education, 10(2), 171-190.	Publication	STEM education/ research	Partnerships/ systemic reform
1100	Johnson, C.C., Kahle, J.B., & Fargo, J.D. (2006). A study of the effect of sustained, whole-school professional development on student achievement in science. Journal of Research in Science Teaching, 44(6), 775-786.	Publication	STEM education/ research	Professional Development
1101	Johnson, C.C., Kahle, J.B., & Fargo, J.D. (2006). Effective teaching results in increased science achievement for all students. Science Education, 91(3), 371-383. ED# EJ60015.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1102	Johnson, E. (2003). Meeting the challenge for sustainability. Bio-Link Connections newsletter. Volume 5, Issue 1.	Publication	STEM education/ research	Project Capacity Building
1103	Jones, M.T. & Eick, C.J. (2007). Providing Bottom-Up Support to Middle School Science Teachers' Reform Efforts in Using Inquiry-Based Kits. Journal of Science Teacher Education.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1104	Jones, M.T., & Eick, C.J. Implementing inquiry kit curriculum: Obstacles, adaptations, and practical knowledge development in two middle school science teachers. Science Education, 91(3), 492-513.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1105	Kanaya, T., Light, D., & Culp, K.M. (2005). Factors Influencing Outcomes from a Technology-Focused Professional Development Program. Journal of Research on Technology in Education, 37(3), 313.	Publication	STEM education/ research	Professional Development

1106	Kansky, B. (nd). Report Summary: Professional Development – Insights from the Local Systemic Change Initiatives. CPRE Policy Report. Philadelphia, PA: Consortium for Policy Research in Education. Available from: http://mathforum.org/kb/servlet/JiveServlet/download/204-1529691-5505263-383119/RSS47.DOC	Report	STEM education/ research	Professional Development
1107	Katzenmeyer, C. & Lawrenz, F. (2006). National Science Foundation Perspectives on the Nature of STEM Program Evaluation. In Huffman, D. & Lawrenz, F. (Eds.), Critical Issues in STEM Evaluation: New Directions for Evaluation, 109. (pp. 7-18).	Publication	STEM evaluation	Evaluation
1109	Kern, A.L, Moore, T.J, & Akillioglu, F.C. (2007, October 10-13). Cooperative Learning: Developing an Observation Instrument for Student Interactions. 37th ASEE/IEEE Frontiers in Education Conference, Milwaukee, WI.	Presentation	Education - general	Evaluation
1110	Khourey-Bowers, C., Dinko, R.L., & Hart, R.G. (2005). Influence of a shared leadership model in creating a school culture of inquiry and collegiality. Journal of Research in Science Teaching, 42(1), 3-24.	Publication	STEM education/ research	Other - Leadership
1111	Kimble, L.L., Yager, R.E. & Yager, S.O. (2006). Success of a Professional-Development Model in Assisting Teachers to Change Their Teaching to Match the More Emphasis Conditions Urged in the National Science Education Standards. Journal of Science Teacher Education, 17(3), 309-322.	Publication	STEM education/ research	Professional Development
1112	Klentschy, M.P., & Maruca, P.M. (2006). The administrator's role in supporting science and literacy in classrooms. In R. Douglas, M.P. Klentschy, & K. Worth. (Ed.), Linking Science and Literacy in the K-8 Classroom (pp. 407-427). Arlington, VA: National Science Teachers Association Press.	Publication	STEM education/ research	Other - Administration
1113	Knapp, A.K. (2006). Activities in algebra: Is GK-12 effective for high school algebra students? Presentation made at the regional meeting of the National Council of Teachers of Mathematics, Chicago, IL.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1114	Krasny, ME. (2005). University K-12 science outreach programs: How can we reach a broad audience? Bioscience, 55(4), 350-359.	Publication	STEM education/ research	Partnerships/ systemic reform

1115	Laursen, S.L. (2007). Getting Unstuck: Strategies for Escaping the Science Standards Straitjacket. The Astronomy Education Review, 5(1), 162-177.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1116	Lawrence Hall of Science. (2007). Database of FOSS Research on FOSS and Ongoing Projects. University of California: author. Available from: http://www.lawrencehallofscience.org/foss/scope/research/search.php	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment
1118	Lawrenz, F. & Post, T. (1999, Spring). Local Systemic Change Initiatives in Science and Mathematics. Research/Practice, 7(1). Minneapolis, MN: Center for Applied Research and Educational Improvement.	Instrument/Tool	STEM education/ research	Professional Development
1119	Lawrenz, F. (2002). Minneapolis Local Systemic Change Initiative Core Evaluation Report: 2001-2002 Final Year. Available from: http://tis.mpls.k12.mn.us/sites/5df1b159-7ce3-4aa3-8e71-8e60a7b98e6c/uploads/LSCeval2002_2.pdf	Report	STEM evaluation	Professional Development
1120	Lawrenz, F. (2007). Review of science education program evaluation. In S. Abell & N.G. Lederman (Eds.), Handbook of Research on Science Education, (pp. 943-954). Mahwah, NJ: Lawrence Erlbaum.	Publication	STEM evaluation	Evaluation
1123	Lawrenz, F., & Huffman, D. (2006). Methodological Pluralism: The Gold Standard of STEM Evaluation. Critical issues in STEM Evaluation. New Directions for Evaluation, 109, 19-34.	Publication	STEM evaluation	Evaluation
1125	Lawrenz, F., Huffman, D. & Gravely, A. (2007). Impact of the collaboratives for excellence in teacher preparation program. Journal of Research in Science Teaching, 44(9), 1348 - 1369.	Publication	STEM education/ research	Professional Development
1131	Lee, M.H., Olson, T.A., & Scribner, J.B. (2006, May). Exploring Alternative Teacher Certification Policy and Practice through an Examination of Novice Science Teachers. Paper presented at the STEM ACT Conference, Arlington, VA.	Report	STEM education/ research	Teacher training/ Pre-service
1132	Lemtrakul, W. (2003). A Development of the Evaluation Method to Assess the Success of a Teacher Training Project Based on Theory – Based Evaluation.	Presentation	Evaluation	Evaluation

1133	Lightbourne, J. (2004, June 15). Memorandum: COV for Teacher Preparation (CETP and STEMTP) Program: COI and Diversity Memo. Washington, DC: Directorate for Education and Human Resources, National Science Foundation. Available from: http://www.nsf.gov/od/oia/activities/cov/ehr/2004/DUE_TEcov.pdf	Report	STEM evaluation	NSF - COV
1134	Lord, B., & Miller, B. (2000, March). Teacher Leadership: An Appealing and Inescapable Force in School Reform? Newton, MA: Education Development Center.	Report	Education - general	Partnerships/ systemic reform
1135	Loucks-Horsley, S., & Matsumoto, C. (1999). Research on Professional Development for Teachers of Mathematics and Science: The State of the Scene. School Science and Mathematics, 99.	Publication	STEM education/ research	Professional Development
1136	Loucks-Horsley, S., Love, N., Stiles, K.E., Mundry, S., & Hewson, P.W. (2003). Designing Professional Development for Teachers of Science and Mathematics, 2nd Edition. Thousand Oaks, CA: Corwin Press.	Publication	STEM education/ research	Professional Development
1137	Lövgren, B. (2005, October 20-21). Platsbesök as method wide evaluation of education. Paper to the Swedish Evaluation Association's conference. [Britta Lövgren, Platsbesök som metod vid utvärdering av utbildning. Paper till Svenska utvärderingsföreningens konferens 20-21 oktober 2005.]	Presentation	Evaluation	Evaluation
1138	Lumpe, A.T., Czerniak, C.M. & Haney, J.J. (1999, June). Supporting the Implementation of Inquiry-based Elementary Science Programs: Setting the Stage for Local Reform. Electronic Journal of Science Education, 3(4).	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1139	MacIsaac, D., Falconer, K., & Zawicki, J. (2005). Using the Reformed Teaching Observation Protocol (RTOP) to facilitate novice teachers in sustaining extraordinary levels of student discourse in introductory physics. Presentation at the American Association of Physics Teachers, Salt Lake City, Utah.	Presentation	STEM evaluation	Teacher training/ Pre-service
1140	MacIsaac, D., Sawada, D., & Falconer, K. (2001, April). Using the Reformed Teaching Observation Protocol (RTOP) as a Catalyst for Self-Reflective Change in Secondary Science Teaching. Paper presented at the American Educational Research Association annual conference, Seattle, Washington.	Presentation	STEM evaluation	Professional Development

1141	Madfes, T.J. & Rosen, R. (2004, November). Follow-up evaluation of MASTEP [Mathematics and Science Teacher Education Program]: Final Report. San Francisco: WestEd.	Report	STEM evaluation	Teacher training/ Pre-service
1142	Mancini, F. (2005, October 19-21). Comparative evaluation of participatory and conventional approaches to impact assessment- A SWOT analysis. Paper presented at the CIGAR Impact Assessment Workshop, CIMMYT Headquarters, Mexico.	Presentation	Evaluation	Evaluation
1143	Mancini, F. (2006. Impact of IPM Farmer Field Schools on the environment, health and livelihoods of cotton growers in Southern India. Doctoral thesis, Biological Farming Systems Group, Wageningen University, The Netherlands.	Publication	Other	Other – Rural Development
1144	Mark, M. M. (2004). From the editor. American Journal of Evaluation, 25(3), 273-275.	Publication	Evaluation	Evaluation
1145	Marshall, J.E. & Bock, S.C. (2006, November). Science teacher retention: What if… CSP Connection, 7(2), 3-4.	Publication	STEM education/ research	Professional Development
1146	McCaffrey, D.F., Hamilton, L.S., Stecher, B.M., Klein, S.P., Bugliari, D., & Robyn, A. (2001, November). Interactions among Instructional Practices, Curriculum, and Student Achievement: The Case of Standards-Based High School Mathematics. Journal for Research in Mathematics Education, 32(5), 493-517.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1147	McClintock, E., O'Brien, G., & Jiang, Z. (2005, Summer). Assessing Teaching Practices of Secondary Mathematics Student Teachers: An Exploratory Cross Case Analysis of Voluntary Field Experiences. Teacher Education Quarterly.	Publication	STEM evaluation	Curriculum/ Instruction/ Assessment
1148	Michaels, S. Shouse, A.W., & Schweingruber, H.A. (2007). Ready, Set, Science!: Putting Research to Work in K-8 Science Classrooms. Washington, DC: The National Academies Press.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1150	Milanowski, T. & Kimball, S. (2006). Ed Leadership & Policy Analysis 703: Evaluating and Supporting Quality Classroom Teaching Course Syllabus. Madison, WI: University of Wisconsin.	Other	Evaluation	Curriculum/ Instruction/ Assessment

1151	Miller, C., Stewart, M., & Townsend, J. (2004, March 14-15). Operational Approaches to the Observation Process. Presentation at the Teacher Preparation PI Conference: NSF CETP, STEMTP, ATE Programs.	Presentation	STEM evaluation	Evaluation
1152	Millsap, M.A., & O'Reilly, F. (nd). Building Evaluation Capacity with Theory-Based Evaluation. Abt Associates, Inc. Used as an example of a successfully funded grant proposal on the NSF REC website. Available from: http://www.ehr.nsf.gov/rec/programs/evaluation/	Other	Evaluation	Evaluation
1153	Minnick, K.F., & McCormick, R. (2005, April 19). DRAFT TxCETP Evaluation Report 2000 - 2004. Albuquerque, NM: Minnick & Associates.	Report	STEM evaluation	Teacher training/ Pre-service
1154	Morrell, P.D., Flick, L., & Wainwright, C. (2004, May). Reform Teaching Strategies Used by Student Teachers. School Science and Mathematics, 104(5), 199-213.	Publication	STEM education/ research	Teacher training/ Pre-service
1155	MSPnet: Math Science Partnerships' Learning Network. (2007) Library: Lessons from a Decade of Mathematics and Science Reform: A Capstone Report for the Local Systemic Change Through Teacher Enhancement Initiative. Available from http://hub.mspnet.org/index.cfm/12760	Electronic Resource	STEM evaluation	Professional Development
1156	Mundry, S. (2005, June). What Experience Has Taught Us About Professional Development. National Network of Eisenhower Regional Consortia and Clearinghouse.	Report	STEM education/ research	Professional Development
1157	National Education Writers Association. (2006, June). Education Reform: No Child Left Behind and Science Education.	Publication	STEM education/ research	Partnerships/ systemic reform
1158	National Science Foundation, Directorate for Education and Human Resources. (2002, January). Program Solicitation: Instructional Materials Development program. Washington, DC: author. http://nsf.gov/pubs/2002/nsf02067/nsf02067.txt	Other	STEM education/ research	NSF - RFP
1159	National Science Foundation. (1998, May 28). Agency information collection activities: Cross Projects Evaluation of the Local Systemic Change Through Teacher Enhancement Program. Federal Register, 63(102). Available from: http://vlex.com/vid/23373036	Other	STEM evaluation	NSF – Renewal solicitation

1160	National Science Foundation. (2006, March). Media Advisory 06-006 Panel Reviews a Decade of Math and Science Education Reform. Available from http://www.nsf.gov/news/news_summ.jsp?cntn_id=106713.	Other	STEM education/ research	NSF – Media advisory
1161	National Science Foundation: Division of Elementary, Secondary and Informal Education (ESIE). (2003, May 11). Response to the Committee of Visitors (COV) Report Presidential Awards for Excellence in Mathematics and Science Teaching (PAEMST) Program. Available from: http://www.nsf.gov/od/oia/activities/cov/ehr/2003/Response_PAEMSTcov.doc.	Report	STEM evaluation	NSF - COV
1162	National Science Foundation; Directorate for Education and Human Resources; Division of Elementary, Secondary, and Informal Education. (2000). Program Announcement: Centers for Learning and Teaching program. Washington DC: National Science Foundation.	Other	STEM education/ research	NSF – program announcement
1163	National Science Teachers Association. (2006, May). NSTA Position Statement: Professional Development in Science Education. Downloaded 10/29/07 from http://www.nsta.org/about/positions/profdev.aspx.	Electronic Resource	STEM education/ research	Professional Development
1164	National Science Teachers Association. (nd). TE-MAT: Teacher Education Materials Project: A Database for K-12 Mathematics and Science Professional Development Providers. Available at: http://www.te-mat.org/ConceptualFramework/references.aspx	Electronic Resource	STEM education/ research	Materials Development
1165	National Sciences Resources Center. (2007). Professional Development Research. Washington, DC: The National Academies. Available from: http://www.nsrconline.org/professional_development/research.html	Electronic Resource	Education - general	Professional Development
1166	National Staff Development Council. (2007, November 20). Learning Communities. Available from: http://64.78.6.92/standards/learningcommunities.cfm?printPage=1&	Electronic Resource	Education - general	Professional Development
1167	Nazar, B. (2006). The logic model: Past, present, and future. The center for applied management website: http://www.appliedmgt.com/eLogic%20Model.htm	Electronic Resource	Evaluation	Evaluation

1168	Needham Public Schools. (2007, June). K-8 Science Program Review Report: Phase II. Needham, MA: author.	Report	STEM evaluation	Partnerships/ systemic reform
1169	Neil Evans, Peter Saflund, Dr. Manjari Wijenaike, John Engman, Kris Madura, & Fran Linhart. [National Workforce Center for Emerging Technologies and CompTIA] (2002). White Paper: IT Security Specialist: Integrating Academic Credentials with IT Professional Certifications. Paper presented at the Cyber-Security Education Workshop sponsored by the National Science Foundation and the American Association of Community Colleges.	Presentation	STEM education/ research	Workforce Development
1170	Nelson, G.D., & Landel, C.C. (2006). A collaborative approach for elementary science. Educational Leadership, 64(4), 72-75.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1171	Neves, P. (2004). NJCATE's National Engineering Technology Education Clearinghouse Facilitates Dissemination and Promotes Sustainability, Key Elements of ATE Projects. National Engineering Technology Education Clearinghouse (NETEC) Network online newsletter. http://www.neteconline.org/news/dec04-sharefindgreatideas.cfm	Electronic Resource	STEM education/ research	Project Capacity Building
1172	North Cascades and Olympic Science Partnership. (2003, June 2). Response to Concerns/Questions. Available from: ncosp.wwu.edu/projectgoals/ReviewResponsesFinal.doc	Report	STEM evaluation	Partnerships/ systemic reform
1173	Paek, P.L, Ponte, E., Sigel, I., Braun, H., & Powers, H. (2004). A Portrait of Advanced Placement Teachers' Practices. College Board Research Report No. 2005-7; ETS RR-05-09. Princeton, NJ: Educational Testing Service.	Report	Education - general	Curriculum/ Instruction/ Assessment
1174	Patton, M., (Ed.) (2006). ATE Centers Impact 2006-2007.Tempe, AZ: Maricopa Community Colleges.	Report	STEM education/ research	Workforce Development
1175	Piburn, M., & Sawada, D. (2001). Reformed Teaching Observation Protocol (RTOP): Reference Manual. ACEPT Technical Report No. IN00-3. Arizona State University: Arizona Collaborative for Excellence in the Preparation of Teachers.	Instrument/Tool	STEM evaluation	Evaluation

1176	Poole, S.J., deGrazia, J.L., & Sullivan, J.F. (1999, November). Assessing K-12 pre-engineering outreach programs. Paper presented at the 29th Annual Frontiers in Education Conference, San Juan: Puerto Rico.	Presentation	STEM evaluation	Workforce Development
1177	Porter, A.C., Garet, M.S., Desimone, L., Yoon, K.S., & Birman, B.F. (2000). Does Professional Development Change Teaching Practice? Results from a Three-Year Study. Washington DC: US Department of Education.	Report	STEM education/ research	Professional Development
1178	Post, T.R., Davis, J.D., Maeda, Y., Edwin, A.C., Andersen, E., Kahan, J.A., & Harwell, M.R. (2004, April). Standards-based Mathematics Curricula and Middle Grade Students' Performance on Standardized Achievement Tests. Minneapolis, MN: University of Minnesota.	Report	STEM education/ research	Curriculum/ Instruction/ Assessment
1179	Potter, P. (2006, July). Experimental housing and urban design: A cluster evaluation methodology for the promotion of innovation in urban development policy. Paper presented at the ENHR Conference, Ljubljana, Slovenia.	Presentation	Evaluation	Other – urban development
1180	Raghavan, K., Cohen-Regev, S., & Strobel, S.A. (2001, December). Student Outcomes in a Local Systemic Change Project. School Science and Mathematics Journal, 101(8), 417-426.	Publication	STEM education/ research	Professional Development
1181	Raine, E. (Unpublished thesis). Where Supply Meets Demand: Women in Diesel Mechanics. University of Kentucky.	Publication	STEM education/ research	Workforce Development
1182	Rebolloso, E., Fernández-Ramírez, B., Cantón, P. & Salvador, C. (2005). "Criterios de calidad de las evaluaciones universitarias" [Criteria of quality of the university evaluations] IV Conferencia de la Sociedad Española de Evaluación.	Presentation	Evaluation	Evaluation
1183	Rhode Island Board of Governors for Higher Education. (2006). PK-16 Advisory Committee on Science Report. Rhode Island: author.	Report	STEM education/ research	Partnerships/ systemic reform
1186	Rose, M.A. (2007, January). ITEDU 694: Curriculum Development in Technology Education: RESOURCES. Syllabus for course taught at Ball State University.	Other	STEM education/ research	Curriculum/ Instruction/ Assessment

1187	Sawada, D., Piburn, M.D., Judson, E., Turley, J., Falconer, K., Benford, R., & Bloom, I. (2002). Measuring Reform Practices in Science and Mathematics Classrooms: The Reformed Teaching Observation Protocol. School Science and Mathematics, 102 (6), 245.	Publication	STEM evaluation	Evaluation
1188	Sawada, D., Piburn, M., Turley, J., Falconer, K., Benford, R., Bloom, I., & Judson, E. (2000, March). Reformed Teaching Observation Protocol (RTOP): Training Guide. ACEPT Technical Report No. IN00-2. Arizona State University: Arizona Collaborative for Excellence in the Preparation of Teachers.	Report	STEM evaluation	Evaluation
1190	Schaetzel, K., Peyton, J.K., & Burt, M. (2007, October). Professional Development for Adult ESL Practitioners: Building Capacity. Washington, DC: Center for Applied Linguistics. Available from: http://www.cal.org/caela/esl_resources/briefs/profdev.html	Electronic Resource	Education - general	Professional Development
1191	Schoen, H.L., Cebulla, K.J,, Finn, K.F., et al. (2003). Teacher variables that relate to student achievement when using a standards-based curriculum. Journal for Research in Mathematics Education, 34(3), 228-259.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1192	Schuster, D., Cobern, W.W., Applegate, B., Schwartz, R., Vellom, P., & Undreiu, A. (2007, October). Assessing pedagogical content knowledge of inquiry science teaching: Developing an assessment instrument to support the undergraduate preparation of elementary teachers to teach science as inquiry. Preprint version. To be published in the Proceedings of the National STEM Assessment Conference on Assessment of Student Achievement, hosted by the National Science Foundation and Drury University, in W	Presentation	STEM evaluation	Evaluation
1193	Science Education Partnership Award. (2006). SEPA-Generated Evaluation Tools from: Brain To Middle Schools. Available from: http://www.ncrrsepa.org/Eval/G_Eval.asp?GrantID=R25RR017315	Electronic Resource	STEM evaluation	Evaluation
1194	Science Storm. (nd). National Science Foundation Award #0119078: A Follow-up Summative Evaluation of the New York City Collaborative for Excellence in Teacher Preparation. Available from: http://www.sciencestorm.com/award/0119078.html	Electronic Resource	STEM evaluation	NSF – grant information

1195	Sciulli, J.A. (2004). Teaching Science Through Inquiry in K-5 Classrooms: Analysis of Change in Practice. Doctoral dissertation: Duquesne University.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1196	Scribner, J.P., Akiba, M., Olson, T.A., Lee, M.H., Nivens, R., & Valentine, E. (2007). Mentoring and teacher practice: Understanding organizational support of standards-based mathematics and science instruction. Paper presented at the Annual Convention of the University Council for Educational Administration, Alexandria, VA	Presentation	STEM education/ research	Professional Development
1197	Seeds of Innovation Website. (nd). Pre-Classroom Observation Interview. Available from: www.iaete.org/challenge/ upload/DIAL_Class_Obs_Protocol.pdf	Instrument/Tool	STEM evaluation	Evaluation
1198	Shapley, K., Sheehan, D., Sturges, K., Caranikas-Walker, F., Huntsberger, B., & Maloney, C. (2006, April). Evaluation of the Texas Technology Immersion pilot: First-year results. Austin: TX: Texas Center for Educational Research.	Report	STEM evaluation	Workforce Development
1199	Shattuck, D. (2006). Technology Implementation in the Classroom. Master's Thesis; North Carolina State University.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1200	Shefner, R. (2007, November). Evaluation without Pain: An Improved Matrix. Presented at the Illinois Mathematics and Science Partnerships Collaborative Meeting #2: Core Needs Assessments and Evaluation Design.	Presentation	STEM evaluation	Evaluation
1202	Shymansky, J.A., Yore, L.D., & Anderson, J.O. (2004). Impact of a school district's science reform effort on the achievement and attitudes of third- and fourth-grade students. Journal of Research in Science Teaching, 41(8), 771-790.	Publication	STEM education/ research	Partnerships/ systemic reform
1205	Sireci, S.G., Zanetti, M.L., Slater, S.C., & Berger, J.B. (2001, September). STEMTEC Evaluation Report For Year 4 (Fall 2000/Spring 2001). Amherst, MA: University of Massachusetts Amherst.	Report	STEM evaluation	Teacher training/ Pre-service
1206	Slater, S.C., Sireci, S.G., Berger, J., Brown, M., O'Meara, K., Hintze, J., & DeCecco, L. (2003, August). STEMTEC II Evaluation Report for Year 1 -- Fall 2002/Spring 2003. Amherst, MA: University of Massachusetts Amherst.	Report	STEM evaluation	Teacher training/ Pre-service

1207	Smith, T.M., & Desimone, L.M. (2003, Spring). Do Changes in Patterns of Participation in Teachers' Professional Development Reflect the Goals of Standards-based Reform? Educational HORIZONS, 119-129.	Publication	Education - general	Professional Development
1208	Smith, T.M., Desimone, L.M., Zeidner, T.L., et al. (2007). Inquiry-oriented instruction in science: Who teaches that way? Educational Evaluation and Policy Analysis, 29(3), 169-199.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1209	Snow-Renner, R., & Lauer, P.A. (2005). McREL Insights Professional Development Analysis. Denver, CO: Mid-continent Research for Education and Learning.	Report	Education - general	Professional Development
1210	Soni, G. (nd). Student Motivation, Attitude, Engagement or Efficacy. Raleigh, NC: Learning in a Technology-Rich Environment, North Carolina State University.	Electronic Resource	STEM education/ research	Other – Student engagement
1211	Sparling, P.B. (2005) Are wellness/fitness programs benefiting participants' movement and mobility in daily life? Quest, 57(1), 162-170.	Publication	Other	Other – Wellness/ fitness programs
1212	St. Olaf College. (nd). Survey and Questionnaire Design. Retrieved from: http://www.stolaf.edu/offices/arp/GrantEvaluations/SurveyQuestionnaireDesign.htm.	Electronic Resource	Evaluation	Evaluation
1213	Stanford University Library and Academic Information Resources. (6/20/05). Sci-Tech Library Newsletter: New e-books and reports. Retrieved from http://www-sul.stanford.edu/depts/swain/nsflibnews/2005/stan050620.html.	Electronic Resource	STEM education/ research	Other – Science/ Tech general
1214	State Educational Technology Directors Association. (nd). ED Tech Instrument Table. Available from: www.setda.org/c/document_library/get_file?folderId=4&name=Examples+of+DDDM.xls	Electronic Resource	STEM education/ research	Evaluation
1215	State of Louisiana. (2007-08). Proposals for LaSIP PD for K-12 Teachers. Available from www.lasip.org/nov28docs/CVCfor2007-08PDProjects.2.ppt	Presentation	STEM education/ research	Professional Development
1216	Stazesky, P.B. (2004, April). Measuring the impact of the quality of mathematics instruction on student achievement during the middle school years: Pilot study. Newark, DE: Delaware Education Research & Development Center, University of Delaware.	Report	STEM education/ research	Curriculum/ Instruction/ Assessment

1217	Supovitz, J.A., & Turner, H.M. (2000). The effects of professional development on science teaching practices and classroom culture. Journal of Research in Science Teaching, 37(9), 963-980.	Publication	STEM education/ research	Professional Development
1218	Taber, M.R., & Quadracci, K. (2006). Building geoscience vocabularies using a data visualization tool (WorldWatcher). In C.A. Manduca and D.W. Mogk, Eds. Earth and Mind: How Geologists Think and Learn about Earth. Special Paper 413. Boulder, CO: The Geological Society of America.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1219	Taylor, J. & Gess-Newsome, J. (2007, January). Exploring Tools and Methods for Measuring Pedagogical Content Knowledge. Presentation at the Annual Meeting of the Association for Science Teacher Education, Clearwater, FL, January 2007.	Presentation	STEM evaluation	Evaluation
1220	The American Association for the Advancement of Science, Northwestern University, The University of Michigan, & Michigan State University. (2006). Preparing the Next Generation of Curriculum Materials Leaders: A Center for the Design, Analysis, and Implementation of Innovative Science Materials. A Proposal to the National Science Foundation for a Center for Learning and Teaching with a Focus on Instructional Materials in Science. Available from: http://ed-web2.educ.msu.edu/CCMS/Webpages/Propo	Other	STEM education/ research	Materials Development
1221	The Art of Teaching Science. (2007, March 6). Reform in Science Teaching, What Does Research Tell Us? Look to a New Cadre. Available from: http://artofteachingscience.org/?cat=6	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment
1222	Council of Chief State School Officers. (2006, March). Selected Research and Resources: Evaluation tools. Available online at: www.ccsso.org/content/PDFs/SelectedResearchandResourcesAnnotatedBib.doc	Electronic Resource	Education - general	Evaluation
1223	The IDEA Partnership: National Association of State Directors of Special Education. (2007, November). Reports, Guides, Instructional Resources, and Opportunities: Math and Science Combined in One Source. Retrieved from http://www.ideapartnership.org/report.cfm?reportid=185.	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment

1224	Tittle, C.K., Flugman, B., Pape, S.M, & Saxman, L. (2001, July). Internal evaluation of the New York Collaborative for Excellence in Teacher Preparation: Final Report. New York: City University of New York, Center for Advanced Study of Education.	Report	STEM evaluation	Teacher training/ Pre-service
1225	Urban Systemic Program. (2004, October 15). Response to the Committee of Visitors. Available from: nsf.gov/od/oia/activities/cov/ehr/2004/Response_USP_COVRpt.doc	Report	STEM evaluation	NSF – COV
1226	Utah State Office of Education. (nd). Science. Available from http://www.schools.utah.gov/curr/sci/SSECC/2006%2009%2021/LSC%20Capstone%20Reportpdf.pdf	Report	STEM education/ research	Professional Development
1227	Van Zoest, & L.R., Stockero, S.L. (2006). The role of curriculum materials in new teachers' practice. In Alatorre, S., Cortina, J.L., Sáiz, M., and Méndez, A.(Eds) (2006). Proceedings of the 28th annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education. Mérida, México: Universidad Pedagógica Nacional.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1228	Vanosdall, R., Klentschy, M., Hedges, L.V, & Weisbaum, K.S. (2007, April). A Randomized Study of the Effects of Scaffolded Guided-Inquiry Instruction on Student Achievement in Science. Paper presented at the Annual Meeting of the American Educational Research Association, Chicago, Illinois.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1229	Wainwright, C. (2007, January 10). Development of a Standards-Based Classroom Observation Instrument. Presentation to the American Association of Physics Teachers conference.	Presentation	STEM evaluation	Evaluation
1230	Wainwright, C. L., Flick, L. B., Morrell, P. (Fall 2003). Development of instruments for assessment of instructional practices in standards-based teaching. Journal of Mathematics and Science: Collaborative Explorations, 6(1), 21-46.	Publication	STEM evaluation	Evaluation
1231	Wainwright, C., Flick, L., Morrell, P.D., & Schepige, A. (2004). Observation of Reform Teaching in Undergraduate Level Mathematics and Science Courses. School Science and Mathematics, 104(7), 322-336.	Publication	STEM education/ research	Evaluation

1232	Wang Ying (2006). Senior U.S. technology education reform plan and the enlightenment to China [translation from Chinese]. Vocational and Technical Education Forum, 23, 53-55.	Publication	STEM education/ research	Partnerships/ systemic reform
1233	Washington State University Wiki "Program Evaluation Resources" http://wiki.wsu.edu/wsuwiki/Program_Evaluation_Resources	Electronic Resource	Evaluation	Evaluation
1234	Weingarth, D. (2006). Perceptions of STARR (Select Teachers as Regional Resources) Teachers Concerning their Professional Development Experience. Doctoral dissertation: University of Missouri-Columbia.	Publication	Education - general	Professional Development
1238	Weiss, I.R. (2006, February). Research on Effective Professional Development. Paper presented at the Council of Chief State School Officers conference: Bridging the Gap between Research & Practice in Professional Development.	Presentation	STEM education/ research	Professional Development
1240	Weiss, I.R., Heck, D.J., Miller, B.A., & Cross, K. (2004, June). Handbook for Enhancing Strategic Leadership in the Math and Science Partnerships. Horizon Research, Inc. and Education Development Center.	Report	STEM education/ research	Partnerships/ systemic reform
1241	Weiss, I.R., Pasley, J.D., Smith, P.S., Banilower, E.R. & Heck, E.R. (2003, May). Looking Inside the Classroom: A Study of K–12 Mathematics and Science Education in the United States. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM education/ research	Curriculum/ Instruction/ Assessment
1244	Wesley, J. (2005). Current characteristics of faculty development in public two-year colleges in Texas. Doctoral dissertation. Texas A&M University.	Publication	Education - general	Professional Development
1245	Westfall, S.A. (2007). The Effect of Professional Development on Physical Education Teachers' Use of Assessment in the Classroom. Doctoral dissertation: Virginia Tech.	Publication	Education - general	Professional Development
1246	National Education Association (2002-2007). What the Research Says. Available from http://www.nea.org/math/research-math.html	Electronic Resource	STEM education/ research	Other – STEM ed variety
1247	Widmer, T., & Frey, K. (2006). Evaluating multilevel network strategies. Zeitschrift fur Evaluation, 2, 287.	Publication	Evaluation	Evaluation

1248	Williams, V.L., Pane, J.F., Tananis, C.A., & Olmsted, S.S. (2005). Math Science Partnership of Southwest Pennsylvania Year Two Evaluation Report. Rand Education Working Paper Series: WR-270-EDU. Santa Monica, CA: Rand Education.	Report	STEM evaluation	Partnerships/ systemic reform
1249	Wood, N.B., Lawrenz, F., Huffman, D., Schultz, M. (2006). Viewing the school environment through multiple lenses: In search of school-level variables tied to student achievement Source: Journal of Research in Science Teaching, 43(3), 237 - 254.	Publication	STEM education/ research	Partnerships/ systemic reform
1250	Xia Jianguo & Liu Xiao Bao. (2007). Application-typed Undergraduate Education: Backdrop and Essential [translation from Chinese]. Research in Higher Education of Engineering, 3.	Publication	STEM education/ research	Workforce Development
1251	Yager, R.E. (2005). Implementing the Changes in Professional Development Envisioned by the National Science Education Standards: Where Are We Eight Years Later? In Exemplary Science: Best Practices in Professional Development. RE Yager, Editor. Arlington, VA: National Science Teachers Association Press.	Publication	STEM education/ research	Professional Development
1252	Yang, H., Shen, J., Cao, H., & Warfield, C. (2004). Multilevel Evaluation Alignment: An Explication of a Four-Step Model. American Journal of Evaluation, 25(4), 493.	Publication	Evaluation	Evaluation
1253	Yates, S.M. (2006). Primary Teachers' Mathematics Beliefs, Teaching Practices and Curriculum Reform Experiences. Paper presented at the Australian Association for Research in Education conference, Adelaide, Australia.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1254	Yore, L.D., & Treagust, D.F. (2006). Current realities and future possibilities: Language and science literacy-empowering research and informing instruction. International Journal of Science Education, 28(2-3), 291-314.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1255	Young, B.J. & Lee, S.K. (2005). The Effects of a Kit-Based Science Curriculum and Intensive Science Professional Development on Elementary Student Science Achievement. Journal of Science Education and Technology, 14(5/6), 471-481.	Publication	STEM education/ research	Professional Development

1256	Berg, D.H., Taylor, J., Hutchinson, N.L., Munby, H., Versnel, J. & Chin, P. (2007). Student assessment in exemplary work-based education programs. Journal of Workplace Learning, 19(4), 209-221.	Publication	Education - general	Workforce Development
1258	Berkeley/Charleston County School District. (2003, January). Report of Fifth Grade Outcome Study, Science for all Students, 2001-2002. Report No. 03-226. Charleston, SC: author.	Report	STEM evaluation	Professional Development
1259	Bickman, L & Gerrard, W. (Spring 2004). Psy315b "Program Evaluation": Course Syllabus. Vanderbilt University.	Other	Evaluation	Evaluation
1260	Bio-link. (nd). Biotechnology Education. Available from: http://www.bio-link.org/skillstandards2.htm.	Electronic Resource	STEM education/ research	Workforce Development
1261	Blank, R.K., de las Alas, N., & Smith, C. (2007, February). Analysis of the Quality of Professional Development Programs for Mathematics and Science Teachers: Findings from a Cross-State Study. Washington DC: Council of Chief State School Officers.	Report	STEM education/ research	Professional Development
1264	Boyle, B., Lamprianou, I, & Boyle, T. (2005). A Longitudinal Study of Teacher Change: What makes professional development effective? Report of the second year of the study. School Effectiveness and School Improvement, 16(1), 1.	Publication	Education - general	Professional Development
1265	Bravo, M. (2004, March). PR-CETP: Evaluation in the Context of Controversies in the Field of Program Evaluation. Presentation to the Teacher Preparation PI Conference: NSF CETP, STEMTP, ATE Programs. Available from: k12s.phast.umass.edu/stemtec/piconf/Proceedings/Papers/Bravo.doc	Presentation	STEM evaluation	Evaluation
1266	Bureau of Justice Assistance, Center for Program Evaluation "Useful Links". http://www.ojp.usdoj.gov/BJA/evaluation/links/index.htm	Electronic Resource	Evaluation	Evaluation
1267	Business-Higher Education Forum. (2007). An American Imperative: Transforming the Recruitment, Retention, and Renewal of our Nation's Mathematics and Science Teaching Workforce. Available from: http://www.bhef.com/solutions/quickfacts_061107.pdf	Report	STEM education/ research	Professional Development

1268	Bybee, R.W., Taylor, J.A., Gardner, A., Van Scotter, P., Powell, J.C., Westbrook, A., & Landes, N. The BSCS 5E Instructional Model: Origins, Effectiveness, and Applications. Colorado Springs, CO: Biological Sciences Curriculum Study.	Report	STEM education/ research	Curriculum/ Instruction/ Assessment
1269	Callow-Heusser, C., Allred, D., Robertson, D., & Sanborn, W. (2005, September 5). Evidence-Based Small-Scale Study Final Report: McDougal Littell Middle School Math Technical Report.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
1270	Callow-Heusser, C., Allred, D., Robertson, D., Borman, G., & Dowling. (2005, July 31). McDougal Littell Evidence-Based Small-Scale Study Final Report: 7th Grade Life Science Technical Report.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
1271	Callow-Heusser, C., Allred, D., Sanborn, W., & Robertson, D. (2005, September 6). Evidence-Based Small-Scale Study Final Report: McDougal Littell Algebra 1, 2nd Edition Technical Report.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
1272	Castellano, M., Stone, J.R., & Stringfield, S. (2005). Earning industry-recognized credentials in high school: Exploring research and policy issues. Journal of Career and Technical Education, 21(2).	Publication	STEM education/ research	Workforce Development
1273	Cavanagh, S. (2006, March 7). NSF Educator-Training Effort Seen as Helpful. Education Week (web only). Available from: http://realizethedream.org/education_news/remote-frame.jsp?base=%5C/education_news/&id=28492117&hostID=28265214 and http://www.edweek.org/ew/articles/2006/03/08/26nsf.h25.html	Publication	STEM education/ research	Professional Development
1274	Center for Applied Research and Educational Improvement. (2006). SSI Impact Study: Project Description. Minneapolis, MN: University of Minnesota. Available from: http://cehd.umn.edu/CAREI/SI/Description.html	Report	STEM evaluation	Partnerships/ systemic reform
1275	Center for Inquiry in Science Teaching and Learning. (nd). Bibliography. Available from: http://cistl.wustl.edu/Bibliography/index.php?ProgramID=0&showall=1	Electronic Resource	STEM education/ research	Curriculum/ Instruction/ Assessment

1276	Desimone, L.M., Le Floch, K.C., Ansell, S., & Taylor, J. (2005, April). Are We Asking the Right Questions on National Surveys about Teacher's Professional Development? Insights from Cognitive Interviews with Teachers (Working Draft). Presented at the Annual Meeting of the American Education Research Association.	Presentation	STEM evaluation	Evaluation
1278	Poole, S.J., deGrazia, J.L., & Sullivan, J.F. (2001, December). Assessing K-12 pre-engineering outreach programs. Journal of Engineering Education.	Publication	STEM evaluation	Workforce Development
1280	National Science Foundation, Directorate for Education and Human Resources, Division of Elementary, Secondary and Informal Education. (2005, September 21). Information Technology Experiences for Students and Teachers (ITEST): Program Solicitation. NSF 05-621. Washington, DC: Author.	Other	STEM education/ research	NSF - RFP
1281	Youth Learn. (2006). Youth & Media Evaluation - Collaborative Research. Newton, MA: Education Development Center. Available from: http://www.youthlearn.org/youthmedia/evaluation/collaborative_research.asp	Electronic Resource	Evaluation	Evaluation
1282	Bergin, K. & Winkler, A. (2007). Challenges and Trade-offs in Measuring the Outcomes of NSF's Math and Science Partnership Program. Presentation at the MSP program regional conference, Pheonix, AZ.	Presentation	STEM evaluation	Evaluation
1283	Zubrowski, B. (2007, December). An Observational and Planning Tool for Professional Development in Science Education. Journal of Science Teacher Education, 18(6), 861-884.	Publication	STEM evaluation	Evaluation
ATE001	Gullickson, A. R., Coryn, C. L. S., & Ritchie, L. A. (2006). Program evaluation. [aka "Project level evaluation practices"] [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #1]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Evaluation
ATE002	Coryn, C. L. S., Gullickson, A. R., & Ritchie, L. A. (2006). Contributors and inhibitors influencing program improvement [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #2]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building

ATE003	Ritchie, L. A., Gullickson, A. R., & Coryn, C. L. S. (2006). Workforce needs assessment [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #3]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Evaluation
ATE004	Coryn, C. L. S., Gullickson, A. R., & Ritchie, L. A. (2006). Contributors and inhibitors influencing articulation agreements [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #4]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Project Capacity Building
ATE005	Ritchie, L. A., Gullickson, A. R., & Coryn, C. L. S. (2006). Challenges and resolutions [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #5]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development
ATE006	Gullickson, A. R., Coryn, C. L. S., & Ritchie, L. A. (2006). Materials development, professional development, and program improvement: Productivity and quality [Advanced Technological Education Program Evaluation Briefing Paper Series, Briefing Paper #6]. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development
ATE018	Gullickson, A. R., & Tressler, G. R. (2002). Recruitment and retention. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 103-129). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Workforce Development
ATE019	Gold, N., & Powe, K. W. (2002). Assessing the impact and effectiveness of professional development. In A. R. Gullickson, F. P. Lawrenz, & N. Keiser (Eds.), The ATE program: Issues for consideration: A monograph (pp. 81-102). Washington, DC: National Science Foundation.	Publication	STEM evaluation	Professional Development
ATE025	Zinser, R. & Hannsen, C. (2006). Improving Access to the Baccalaureate: Articulation Agreements and the National Science Foundation's Advanced Technological Education Program. Community College Review, 34(1), 27-43.	Publication	STEM evaluation	Workforce Development

ATE028	Zinser, R., & Lawrenz, F. P. (2004). New roles to meet industry needs: A look at the Advanced Technological Education program. Journal of Vocational Education Research, 29(2), 85-100.	Publication	STEM evaluation	Workforce Development
ATE029	Lawrenz, F. P., Keiser, N., & LaVoie, B. (2003). Sustaining innovation in technological education. Community College Review, 30(4), 1-14.	Publication	STEM education/ research	Project Capacity Building
ATE030	Lawrenz, F. P., Keiser, N,.& LaVoie, B. (2003) Evaluative site visits: A methodological review. American Journal of Evaluation, 24(3), 341-352.	Publication	Evaluation	Evaluation
ATE032	Gullickson, A. R., Wingate, L., Lawrenz, F. A., & Coryn, C. L. S. (2006). The National Science Foundation's Advanced Technology Education Program: Final evaluation report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Workforce Development
ATE037	Appleton, J. J., & Lawrenz, F. P. (2005). ATE materials outcome comparison: Technical report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development
ATE038	Appleton, J. J., & Lawrenz, F. P. (2004). ATE materials development process report. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development
ATE039	Keiser, J., Lawrenz, F. P., & Appleton, J. J. (2004). Evaluation of materials produced by the ATE program. Kalamazoo, MI: Western Michigan University, The Evaluation Center.	Report	STEM evaluation	Materials Development
ATE062	Migotsky, C., & Stake, R. (2001). An evaluation of an evaluation: CIRCE's metaevaluation of the ATE program evaluation. Urbana-Champaign, IL: University of Illinois at Urbana-Champaign.	Report	STEM evaluation	Evaluation
ATE087	Hanssen, C. E., & Coryn, C. L. S. (2004, November). Theory-driven evaluations of multi-site programs. Paper presented at the meeting of the American Evaluation Association, Atlanta, GA.	Presentation	Evaluation	Evaluation
CETP001	Lawrenz, F., Gravely, A., and Ooms, A. (2006). Perceived helpfulness and amount of use of technology in science and mathematics classes at different grade levels. School Science and Mathematics, 106(3), 133-139.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment

CETP007	Lawrenz, F., Huffman, D., Appledoorn, K., & Michlin, M. (2002, February). CETP Core Evaluation: Field Test Results. Minneapolis, MN: Center for Applied Research and Educational Improvement, University of Minnesota.	Report	STEM evaluation	Evaluation
CETP010	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. K-12 Surveys Handbook, (Jan, 2001) revised September, 2002	Instrument/Tool	STEM evaluation	Evaluation
CETP011	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. Higher Education Survey Handbook, January, 2001. Revised September 2002	Instrument/Tool	STEM evaluation	Evaluation
CETP012	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. Classroom Observation Handbook, January, 2001 Revised September 2002	Instrument/Tool	STEM evaluation	Evaluation
CETP013	Lawrenz, F., Huffman, D., & Appeldoorn, K. (2002, September). Handbook for Evaluation of Activities and Assessments. Minneapolis, MN: Collaboratives for Excellence in Teacher Preparation, University of Minnesota.	Instrument/Tool	STEM evaluation	Evaluation
CETP015	Lawrenz, F. (2003, June). Differential Effects of Participatory Evaluation in a National Multi-site Program Evaluation. Presentation at the Canadian Evaluation Society national meeting, Vancouver, Canada.	Presentation	Evaluation	Evaluation
LSC001	Banilower, E. R. & Germuth A. A. (2004). Structural equation models for LSC teacher questionnaire and classroom observation data. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Evaluation
LSC003	Banilower, E. R., Boyd, S. E., Pasley, J. D., & Weiss, I. R. (2006). Lessons from a decade of mathematics and science reform: A capstone report for the local systemic change through teacher enhancement initiative. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC006	Banilower, E.R. (2002). Results of the 2001–2002 study of the impact of the local systemic change initiative on student achievement in science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC007	Bowes, A. S. & Banilower, E. R. (2004). LSC classroom observation study: An analysis of data collected between 1998 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC009	Boyd, S. E., Banilower, E. R., Pasley, J. D., & Weiss, I. R. (2003). Progress and pitfalls: A cross-site look at local systemic change through teacher enhancement. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development

LSC012	Germuth, A. A. & Banilower, E. R. (2004). Results of the 2002–03 study of the impact of the local systemic change initiative on student achievement in science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC014	Heck, D. J. & Crawford, R. A. (2004). LSC teacher questionnaire study: A longitudinal analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC026	Shimkus, E.S., & Banilower, E.R. (2004, June). Local Systemic Change: Comparison of Teachers' and Principals' Perceptions of Factors Influencing Instruction. Chapel Hill, NC: Horizon Research Inc.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
LSC027	Simpson, M. A. & Banilower, E. R. (2004). LSC principal questionnaire study: A longitudinal analysis of data collected between 1997 and 2003. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC028	Simpson, M. A. & Banilower, E. R. (2004). Results of the 2003–04 Study of the Impact of the Local Systemic Change Initiative on Student Achievement in Science. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC031	Weiss, I. R., Arnold, E. E., Banilower, E. R., & Soar, E. H. (2001). Local systemic change through teacher enhancement: Year six cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC033	Weiss, I. R., Banilower, E. R., Crawford, R. A., & Overstreet, C.M. (2003). Local systemic change through teacher enhancement: Year eight cross-site report. Chapel Hill, NC: Horizon Research, Inc.	Report	STEM evaluation	Professional Development
LSC043	Banilower, E. R., Heck, D. J., & Weiss, I. R. (2007). Can professional development make the vision of the standards a reality? The impact of NSF's local systemic change through teacher enhancement initiative. Journal of Research in Science Teaching, 44(3), 375–395.	Publication	STEM education/ research	Professional Development
LSC045	Weiss, I. R. & Pasley, J. D. (2006, March). Scaling up instructional improvement through teacher professional development: Insights from the local systemic change initiative. CPRE Policy Briefs, RB-44.	Publication	STEM education/ research	Professional Development
LSC046	Weiss, I.R., & Pasley, J.D. (2007). Teaching math and science - Improving instruction through local systemic change initiatives. Phi Delta Kappan, 88(9), 669-675.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment

LSC065	Weiss, Iris R. (2005, May). Findings from the LSC. Presentation at the CCSSO Conference.	Presentation	STEM evaluation	Professional Development
LSC066	Weiss, Iris R. (2005, November). High Quality Mathematics and Science Instruction: Where do we stand? How do we move forward? Presentation to Pittsburgh Public Schools.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
LSC096	Horizon Research. (1997, March). LSC Core Update. Chapel Hill, NC: author.	Report	STEM evaluation	Professional Development

Appendix I: Citing Authors

Author Name	N	%
633 Study Group	1	0.4
Abell et al.	2	0.7
Access My Library	1	0.4
ACEPT	6	2.1
Alabama Education	1	0.4
American Association for the Advancement of Science	1	0.4
American Society for Biochemistry and Molecular Biology	1	0.4
Art of Teaching Science	1	0.4
Asset Inc. Teacher Enhancement	4	1.4
Astrophysics Data System	1	0.4
ATE Center Evaluation	1	0.4
ATE Evaluation	18	6.4
ATE other evaluation	1	0.4
Basista & Mathews	1	0.4
Beasley	1	0.4
Berg et al.	1	0.4
Bickman & Gerrard	1	0.4
Bio-link	2	0.7
Blank et al.	1	0.4
Boyle et al.	1	0.4
Bureau of Justica Assistance	1	0.4
Business-Higher Education Forum	1	0.4
Bybee et al.	1	0.4
Callow-Heusser	3	1.1
CAREI	1	0.4
Castellano et al.	1	0.4
Cavanagh	1	0.4
CCSSO	3	1.1
Center for Inquiry in Science Teaching and Learning	1	0.4
Center for Science, Mathematics, and Engineering Education	1	0.4
CETP-PA	1	0.4
CETP Core Evaluation	8	2.9
China Medical College	1	0.4
Chrispeels	1	0.4
Committee on Science Learning	1	0.4
Connecticut Education	1	0.4
Coryn	1	0.4
CPRE Policy Brief	1	0.4

D'Ambrosio et al.	1	0.4
D'Avanzo et al.	1	0.4
Davidson	1	0.4
Desimone el al.	1	0.4
Desimone et al.	6	2.1
Director of the Science, Mathematics, and Technology Education program at Western Washington Univers	1	0.4
Dirr	1	0.4
Donovan & Pellegrino	1	0.4
Duggan-Haas	1	0.4
Education Commission of the States	2	0.7
Education Research Associates	1	0.4
Ellis et al.	1	0.4
Eun & Heining-Boynton	1	0.4
European Community	1	0.4
Evaluation of Professional Development Projects	1	0.4
Ferrini-Mundy	1	0.4
Fletcher et al.	1	0.4
Fouts et al.	1	0.4
Fraser & Kahle	1	0.4
Georgia Tech	1	0.4
Gilbert Systemic Science Plan	1	0.4
Goliath	1	0.4
Good	1	0.4
Goodyear	1	0.4
Goss	1	0.4
Goss & Newcombe	1	0.4
Grable & Park	1	0.4
Grant	1	0.4
Hake	1	0.4
Hamilton et al.	3	1.1
Hartshorne	1	0.4
Harvard GSE	1	0.4
Heller et al.	2	0.7
Heywood and Middleton Primary Care Trust	1	0.4
High & Nemes	2	0.7
HighBeam Research	1	0.4
Huang	1	0.4
Huffman	3	1.1
Illinois Math-Science Partnership	1	0.4
Implementing Investigations in Mathematics	2	0.7
Indiana Mathematics Initiative	1	0.4
International Society for Computational Biology	1	0.4
Inverness Research	1	0.4

Iona	1	0.4
Jeanpierre	1	0.4
Johnson CC	4	1.4
Jones & Eick	2	0.7
Kanaya	1	0.4
Kern et al.	1	0.4
Kimble et al.	1	0.4
KITES: Kits in Teaching Elementary Science	1	0.4
Knapp	1	0.4
Krasny	1	0.4
Laursen	1	0.4
Lawrenz	6	2.1
Lemtrakul	1	0.4
Looking Inside the Classroom	1	0.4
Loucks-Horsley et al.	2	0.7
Louisiana Education	1	0.4
Lovgren	1	0.4
LSC Core Evaluation	21	7.5
MacIssac et al.	1	0.4
Mancini	2	0.7
Mark	1	0.4
Marshall & Bock	1	0.4
MASTEP	1	0.4
Math Science Partnership of Southwest Pennsylvania	1	0.4
McClintock et al.	1	0.4
Michaels et al.	1	0.4
Mid-continent Research for Education and Learning	1	0.4
Milanowski & Kimball	1	0.4
Millsap & O'Reilly	1	0.4
Minneapolis Public Schools Systemic Change in Science Initiative	1	0.4
Missouri ATCP project	2	0.7
MSP	2	0.7
MSP Net	1	0.4
Mundry	1	0.4
National Association of State Directors of Special Education	1	0.4
National Education Association	1	0.4
National Education Writers Association	1	0.4
National Science Teachers Association	2	0.7
National Sciences Resources Center	1	0.4
National Staff Development Council	1	0.4
National Workforce Center for Emerging Technologies	1	0.4
Nazar	1	0.4
Needham Public Schools	1	0.4

Neves	1	0.4
North Cascades and Olympic Science Partnership	1	0.4
NSF	2	0.7
NSF, EHR	4	1.4
NSF, ESIE	1	0.4
NYCETP	3	1.1
O-TEC	1	0.4
OCEPT	5	1.8
Paek et al.	1	0.4
Poole et al.	2	0.7
Potter	1	0.4
PR-CETP	1	0.4
PRIME: Pittsburgh Reform in Mathematics Education	1	0.4
Project Inquiry: Building a Presence for Science in the Lowcountry	1	0.4
Raine	1	0.4
Reaching Every Teacher: A Systemic Approach to Increased Student Achievement K-12	1	0.4
Rebolloso	1	0.4
Rhode Island Education	1	0.4
Rose	1	0.4
Schaetzel et al.	1	0.4
Schoen et al.	1	0.4
Schuster et al.	1	0.4
Science Education Partnership Award	1	0.4
Science PALs	1	0.4
Sciulli	1	0.4
Seeds of Innovation	1	0.4
Shapley et al.	1	0.4
Shattuck	1	0.4
Soni	1	0.4
Sparling	1	0.4
St. Olaf College	1	0.4
Stanford University	1	0.4
State Educational Technology Directors Association	1	0.4
STEMTEC	2	0.7
Students Using Mathematics Successfully (SUMS)	1	0.4
Supovitz& Turner	1	0.4
Systemic Reform of Mathematics K-5 for Virginia	1	0.4
Taber & Quadracci	1	0.4
TAPESTRIES	1	0.4
Taylor & Gess-Newsome	1	0.4
TxCETP	1	0.4
UC Berkeley	1	0.4
Urban Systemic Program	1	0.4

Utah Education	1	0.4
Valle Imperial - Project in Science	1	0.4
Van Zoest & Stockero	1	0.4
Wang	1	0.4
Washington State University	1	0.4
Weingarth	1	0.4
Weiss	1	0.4
Wesley	1	0.4
Westfall	1	0.4
Widmer	1	0.4
Xia & Liu	1	0.4
Yager	1	0.4
Yang et al.	1	0.4
Yates	1	0.4
Youth Learn	1	0.4
Zubrowski	1	0.4
Total	280	100%